MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE EMQ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic JK flip-flop module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.342 failures per million operating hours for class I or 3.344 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EMQ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/264-1 for class I and M28787/264-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 3 to 18 V ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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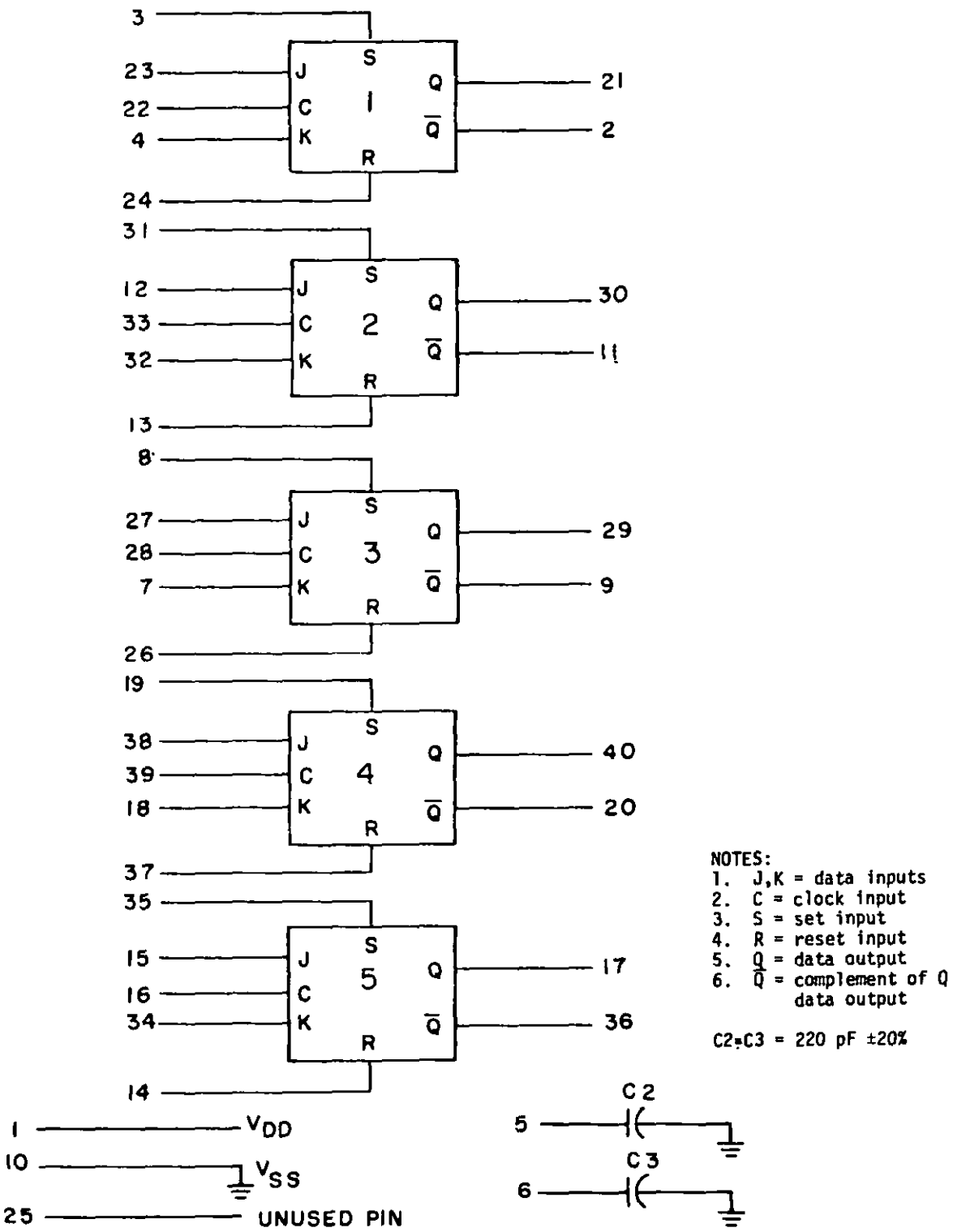


FIGURE 1. EMQ module function diagram, JK flip-flop.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B25	Negative input clamp diode voltage	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C25	Positive input clamp diode voltage	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D7	Functional	VI	X	← See table VI →										
E1-E10	Output voltage (low level), (I_{OL} = 0)	VII	X	0	50	0	50	0	50	0	50	0	50	mV
F1-F10	Output voltage (high level), (I_{OH} = 0)	VII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
G1-G10	Output voltage (low level), V_{DD} = 5 V	VIII	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H10	Output voltage (high level), V_{DD} = 5 V	IX	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
J1-J10	Output voltage (low level), V_{DD} = 15 V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
K1-K10	Output voltage (high level), V_{DD} = 15 V	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
L1-L25	Input current (low level)	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
M1-M25	Input current (high level)	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
N1	Power supply current (low level)	XIV	X	0	60	0	60	0	590	0	60	0	965	µA
N2	Power supply current (high level)	XIV	X	0	60	0	60	0	590	0	60	0	965	µA
P1-P10	Transition time, t_{TLH}, C_L = 50 pF, V_{DD} = 5 V	XV	S	36	360	36	360	40	400	27	275	44	425	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P11-P20	Transition time, t_{TLH}, C_L = 150 pF, V_{DD} = 5 V	XV	Q	84	840	84	840	93	930	64	640	99	967	ns
Q1-Q10	Transition time, t_{THL}, C_L = 50 pF, V_{DD} = 5 V	XV	S	20	200	20	200	22	220	15	150	24	235	ns
Q11-Q20	Transition time, t_{THL}, C_L = 150 pF, V_{DD} = 5 V	XV	Q	68	680	68	680	75	750	51	515	80	802	ns
R1-R10	Propagation delay time, t_{PLH}, (clock to output), C_L = 50 pF, V_{DD} = 5 V	XVI	S	35	350	35	350	38	385	26	265	41	412	ns
R11-R20	Propagation delay time, t_{PLH}, (clock to output), C_L = 150 pF, V_{DD} = 5 V	XVI	Q	55	550	55	550	61	610	42	420	65	650	ns
S1-S10	Propagation delay time, t_{PHL}, (clock to output), C_L = 50 pF, V_{DD} = 5 V	XVI	S	35	350	35	350	38	385	26	265	41	412	ns
S11-S20	Propagation delay time, t_{PHL}, (clock to output), C_L = 150 pF, V_{DD} = 5 V	XVI	Q	55	550	55	550	61	610	42	420	64	650	ns
T1-T5	Propagation delay time, t_{PLH}, (set to output), C_L = 50 pF, V_{DD} = 5 V	XVII	S	35	350	35	350	38	385	26	265	41	412	ns
T6-T10	Propagation delay time, t_{PLH}, (reset to output), C_L = 50 pF, V_{DD} = 5 V	XVII	S	45	450	45	450	50	500	34	340	53	531	ns
T11-T15	Propagation delay time, t_{PLH}, (set to output), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	55	550	55	550	61	610	42	420	64	650	ns
T16-T20	Propagation delay time, t_{PLH}, (reset to output), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	65	650	65	650	72	720	49	495	77	767	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U1-U5	Propagation delay time, t_{PHL}, (set to output), C_L = 50 pF, V_{DD} = 5 V	XVII	S	35	350	35	350	38	385	26	265	40	412	ns
U6-U10	Propagation delay time, t_{PHL}, (reset to output), C_L = 50 pF, V_{DD} = 5 V	XVII	S	45	450	45	450	50	500	34	340	53	531	ns
U11-U15	Propagation delay time, t_{PHL}, (set to output), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	55	550	55	550	61	610	42	420	64	650	ns
U16-U20	Propagation delay time, t_{PHL}, (reset to output), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	65	650	65	650	72	720	49	495	77	767	ns
V1-V10	Transition time, t_{TLH}, C_L = 50 pF, V_{DD} = 10 V	XVIII	Q	18	180	18	180	20	200	13	135	22	212	ns
W1-W10	Transition time, t_{THL}, C_L = 50 pF, V_{DD} = 10 V	XVIII	Q	10	100	10	100	11	110	7	75	12	120	ns
X1-X10	Propagation delay time, t_{PLH}, (clock to output), C_L = 50 pF, V_{DD} = 10 V	XIX	Q	15	150	15	150	16	165	11	115	18	177	ns
Y1-Y10	Propagation delay time, t_{PHL}, (clock to output), C_L = 50 pF, V_{DD} = 10 V	XIX	Q	15	150	15	150	16	165	11	115	18	177	ns
Z1-Z5	Propagation delay time, t_{PLH}, (set to output), C_L = 50 pF, V_{DD} = 10 V	XX	Q	15	150	15	150	16	165	11	115	18	177	ns
Z6-Z10	Propagation delay time, t_{PLH}, (reset to output), C_L = 50 pF, V_{DD} = 10 V	XX	Q	20	200	20	200	22	220	15	150	24	235	ns
AA1-AA5	Propagation delay time, t_{PHL}, (set to output), C_L = 50 pF, V_{DD} = 10 V	XX	Q	15	150	15	150	16	165	11	115	18	177	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA6-AA10	Propagation delay time, t_{PHL}, (reset to output), C_L = 50 pF, V_{DD} = 10 V	XX	Q	20	200	20	200	22	220	15	150	24	235	ns
BB1-BB10	Transition time, t_{TLH}, C_L = 50 pF, V_{DD} = 15 V	XXI	Q	13	130	13	130	14	145	10	100	16	155	ns
CC1-CC10	Transition time, t_{THL}, C_L = 50 pF, V_{DD} = 15 V	XXI	Q	8	80	8	80	9	90	6	60	10	96	ns
DD1-DD10	Propagation delay time, t_{PLH}, (clock to output), C_L = 50 pF, V_{DD} = 15 V	XXII	Q	10	100	10	100	11	110	7	75	12	120	ns
EE1-EE10	Propagation delay time, t_{PHL}, (clock to output), C_L = 50 pF, V_{DD} = 15 V	XXII	Q	10	100	10	100	11	110	7	75	12	120	ns
FF1-FF5	Propagation delay time, t_{PLH}, (set to output), C_L = 50 pF, V_{DD} = 15 V	XXIII	Q	10	100	10	100	11	110	7	75	12	120	ns
FF6-FF10	Propagation delay time, t_{PLH}, (reset to output), C_L = 50 pF, V_{DD} = 15 V	XXIII	Q	15	150	15	150	16	165	11	115	18	177	ns
GG1-GG5	Propagation delay time, t_{PHL}, (set to output), C_L = 50 pF, V_{DD} = 15 V	XXIII	Q	10	100	10	100	11	110	7	75	12	120	ns
GG6-GG10	Propagation delay time, t_{PHL}, (reset to output), C_L = 50 pF, V_{DD} = 15 V	XXIII	Q	15	150	15	150	16	165	11	115	18	177	ns
HH1-HH10	Maximum shift frequency (TA)	XXIV	S	1200	1465	1200	1465	1200	1465	1200	1465	1200	1465	ns
JJ	Power supply filter capacitance	XXV	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
KK1,KK2	C2 and C3 capacitance	XXVI	X	176	264	176	264	176	264	176	264	176	264	pF
LL1-LL25	Input capacitance	XXVII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through LL shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through LL in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through K and N through KK shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. For the transient thermal test, use the same test conditions as the maximum class temperature conditions.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition JJ, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through KK shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through LL shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table 1.

	Class I	Class II	Unit
Recommended supply voltage	3 to 18	3 to 18	V dc
Quiescent maximum power dissipation (15 V dc operation) (outputs not loaded)	18	18	mW

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	Class I	Class II	Unit
Input voltage			
Low level (V_{DD} = 5.0 V)			
At -55°C to 85°C	0.0 *min*, 1.5 *max*	0.0 *min*, 1.5 *max*	V
Low level (V_{DD} = 15.0 V)			
At -55°C to 85°C	0 *min*, 4 *max*	0 *min*, 4 *max*	V
High level (V_{DD} = 5.0 V)			
at -55°C to 85°C	3.5 *min*, V_{DD} *max*	3.5 *min*, V_{DD} *max*	V
High level (V_{DD} = 15.0 V)	11 min, V_{DD} max	11 min, V_{DD} max	V
Input current			
Low level (V_{IL} = 0 V, V_{DD} = 18 V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18 V, V_{DD} = 18 V)	0 min, 200 max	0 min, 200 max	nA
Output voltage			
Low level (V_{DD} = 5.0 V)			
At -55°C to 85°C (I_{OL} = 0)	0 min, 50 max	0 min, 50 max	mV
At -55°C (I_{OL} = 0.64 mA)	0 min, 400 max	0 min, 400 max	mV
At 0°C and 25°C (I_{OL} = 0.51 mA)	0 min, 400 max	0 min, 400 max	mV
At 60°C (I_{OL} = 0.44 mA)	0 min, 400 max	0 min, 400 max	mV
At 85°C (I_{OL} = 0.36 mA)	0 min, 400 max	0 min, 400 max	mV
High level (V_{DD} = 5.0 V)			
At -55°C to 85°C (I_{OH} = 0)	4.95 min, 5.05 max	4.95 min, 5.05 max	V
At -55°C (I_{OH} = -0.64 mA)	4.6 min, 5.0 max	4.6 min, 5.0 max	V
At 0°C and 25°C (I_{OH} = -0.51 mA)	4.6 min, 5.0 max	4.6 min, 5.0 max	V
At 60°C (I_{OH} = -0.44 mA)	4.6 min, 5.0 max	4.6 min, 5.0 max	V
At 85°C (I_{OH} = -0.36 mA)	4.6 min, 5.0 max	4.6 min, 5.0 max	V
Low level (V_{DD} = 15.0 V)			
At -55°C to 85°C (I_{OL} = 0)	0.0 min, 1.5 max	0.0 min, 1.5 max	V
High level (V_{DD} = 15.0 V)			
At -55°C to 85°C (I_{OH} = 0)	13.5 min, 15.0 max	13.5 min, 15.0 max	V
Transition time			
V_{DD} = 5 V			
t_{TLH} (C_L = 50 pF)	36 min, 400 max	27 min, 425 max	ns
t_{TLH} (C_L = 150 pF)	84 min, 930 max	64 min, 967 max	ns
t_{THL} (C_L = 50 pF)	20 min, 220 max	15 min, 235 max	ns
t_{THL} (C_L = 150 pF)	68 min, 750 max	51 min, 802 max	ns
V_{DD} = 10 V			
t_{TLH} (C_L = 50 pF)	18 min, 200 max	13 min, 212 max	ns
t_{THL} (C_L = 50 pF)	10 min, 110 max	7 min, 120 max	ns
V_{DD} = 15 V			
t_{TLH} (C_L = 50 pF)	13 min, 145 max	10 min, 155 max	ns
t_{THL} (C_L = 50 pF)	8 min, 90 max	6 min, 96 max	ns
Propagation delay time			
V_{DD} = 5 V			
t_{PLH}			
Clock to output (C_L = 50 pF)	35 min, 385 max	26 min, 412 max	ns
(C_L = 150 pF)	55 min, 610 max	42 min, 650 max	ns
Set to output (C_L = 50 pF)	35 min, 385 max	26 min, 412 max	ns
(C_L = 150 pF)	55 min, 610 max	42 min, 650 max	ns
Reset to output (C_L = 50 pF)	45 min, 500 max	34 min, 531 max	ns
(C_L = 150 pF)	65 min, 720 max	49 min, 767 max	ns

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	Class I	Class II	Unit
t_{PHL}			
Clock to output (C_L = 50 pF)	35 min, 385 max	26 min, 412 max	ns
(C_L = 150 pF)	55 min, 610 max	42 min, 650 max	ns
Set to output (C_L = 50 pF)	35 min, 385 max	24 min, 412 max	ns
(C_L = 150 pF)	55 min, 610 max	42 min, 650 max	ns
Reset to output (C_L = 50 pF)	45 min, 500 max	34 min, 531 max	ns
(C_L = 150 pF)	65 min, 720 max	49 min, 767 max	ns
V_{DD} = 10.0 V (C_L = 50 pF)			
t_{PLH}			
Clock to output	15 min, 165 max	11 min, 177 max	ns
Set to Q or Reset to $\overline{Q}$	15 min, 165 max	11 min, 177 max	ns
t_{PHL}			
Clock to output	15 min, 165 max	11 min, 177 max	ns
Set to Q or Reset to $\overline{Q}$	15 min, 165 max	11 min, 177 max	ns
V_{DD} = 15.0 V (C_L = 50 pF)			
t_{PLH}			
Clock to output	10 min, 110 max	75 min, 120 max	ns
Set to Q or Reset to $\overline{Q}$	10 min, 110 max	75 min, 120 max	ns
t_{PHL}			
Clock to output	10 min, 110 max	75 min, 120 max	ns
Set to Q or Reset to $\overline{Q}$	10 min, 110 max	75 min, 120 max	ns
Clock input			
Frequency (V_{DD} = 5.0 V, toggle mode)	1.5 min	1.5 min	MHz
Pulse width	330 max	330 max	ns
Set or Reset pulse width	250 max	250 max	ns
Input times			
Setup time (V_{DD} = 5.0 V)	200 max	200 max	ns
Hold time (V_{DD} = 5.0 V)	0 max	0 max	ns

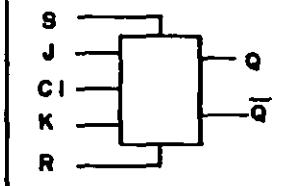
* 6.5 Module function. This module contains five independent CMOS JK flip-flops and two 220 pF capacitors with one side connected to V_{SS}, for external use.

6.5.1 Input to output relationship. Each flip-flop has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship for each JK flip-flop.

INPUTS						OUTPUTS *	
CI †	J	K	S	R	Q_N ‡	Q_{N+1}	$\overline{Q_{N+1}}$
↓	H	X	L	L	L	H	L
↓	X	L	L	L	H	H	L
↓	L	X	L	L	L	L	H
↓	X	H	L	L	H	L	H
↑	X	X	L	L	X	Q_n	$\overline{Q_n}$
X	X	X	H	L	X	H	L
X	X	X	L	H	X	L	H
X	X	X	H	H	X	H	H

X = DON'T CARE
† = LEVEL CHANGE
‡ = PRESENT STATE
* = NEXT STATE
H = HIGH LEVEL
L = LOW LEVEL



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6.5.2 Paralleling outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

a. 2 and 3
b. 4 and 5

6.5.3 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid buildup of static charge.
b. Soldering iron tips, metal parts of fixture and tools, and handling facilities should be grounded.
c. Signals should not be applied to the inputs while the device power supply is off.
d. Input voltages greater than the supply (V_{DD}-V_{SS}) should not be applied as this will forward bias the protection diodes.
e. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

6.5.4 Dynamic power dissipation.

The dynamic power dissipation can be approximated by the formula

$$P_{Total} = (C\ V_{DD}^2\ f)\ N + \text{quiescent power dissipation}$$

where

C = load capacitance.
V_{DD} = supply voltage.
f = operating frequency.
N = number of outputs.

6.6 CAUTION: This caution applies to each test figure. Failure to comply with the following procedures will result in irreversible module damage:

All test equipment must be grounded.
V_{DD} must be applied before applying input signals or condition voltages.
All input signals and conditioning voltages must be removed before V_{DD} is disconnected.

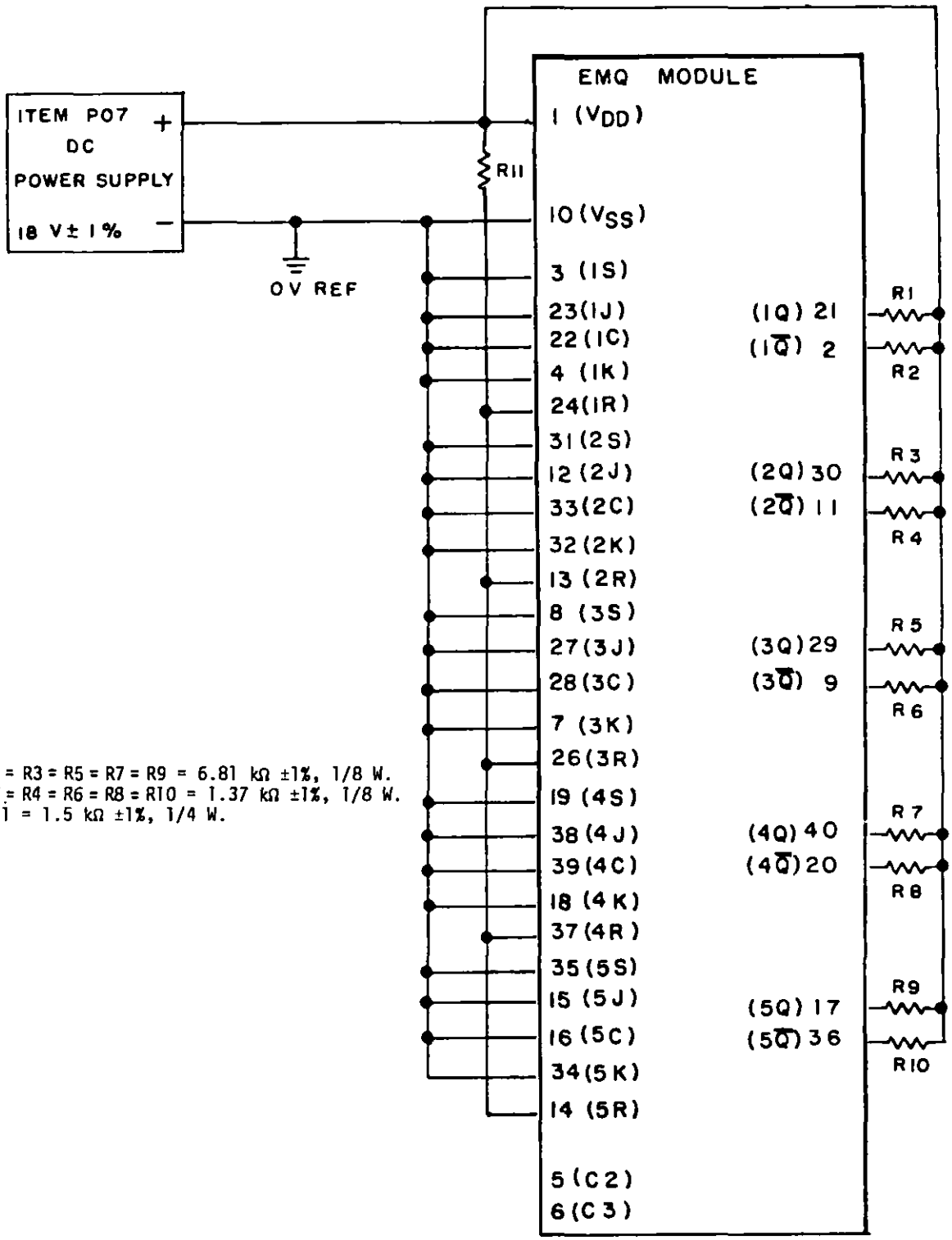
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
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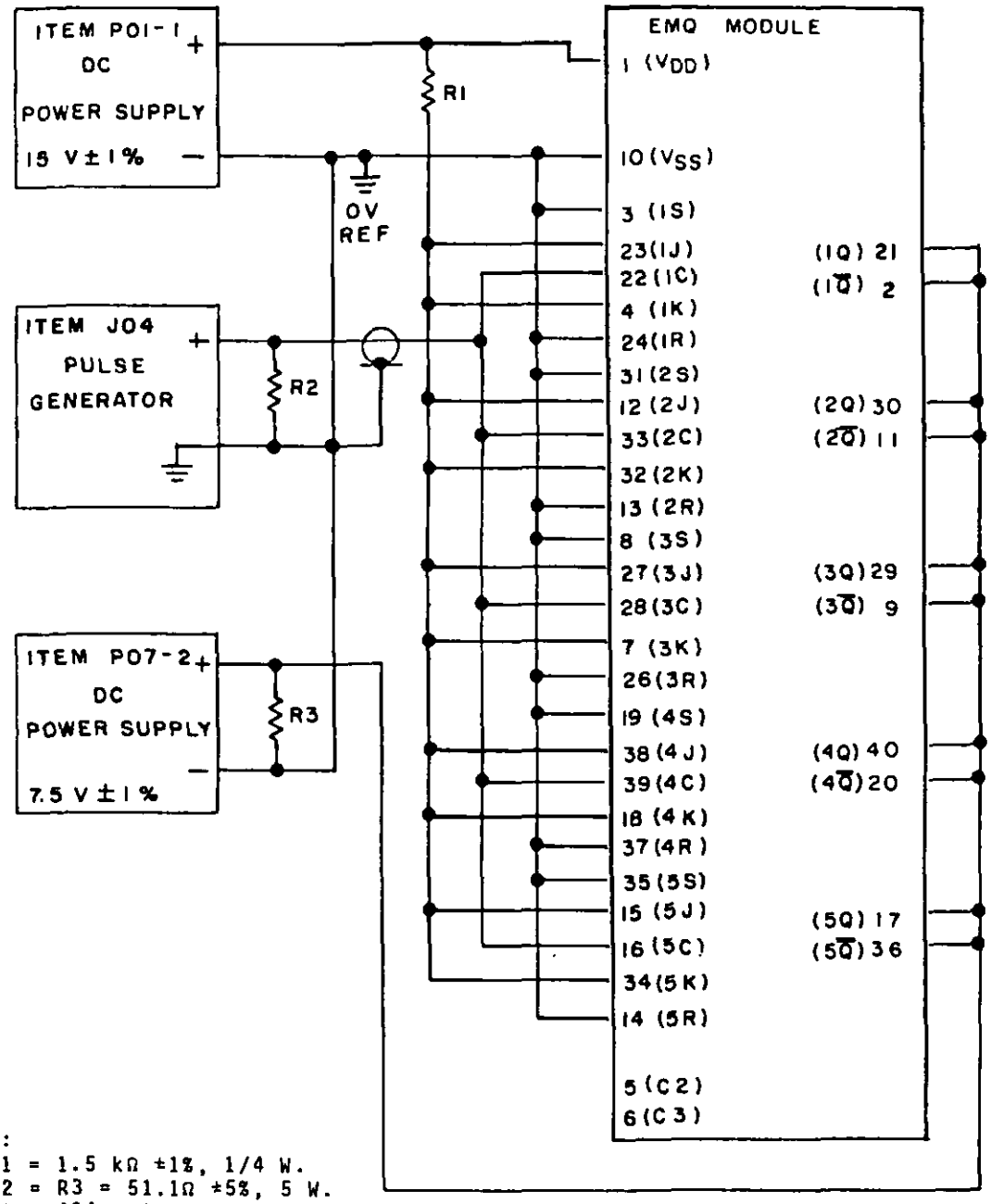
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NOTES:
1. R1 = R3 = R5 = R7 = R9 = 6.81 kΩ ±1%, 1/8 W.
2. R2 = R4 = R6 = R8 = R10 = 1.37 kΩ ±1%, 1/8 W.
3. R11 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/4 W.
2. R2 = R3 = 51.1Ω ±5%, 5 W.
3. Item J04 setup:
   PA = 15 V ±1%.
   PW = 1.0 µs ±200 ns.
   Frequency = 500 kHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.

* FIGURE 3. Component temperature test circuit.

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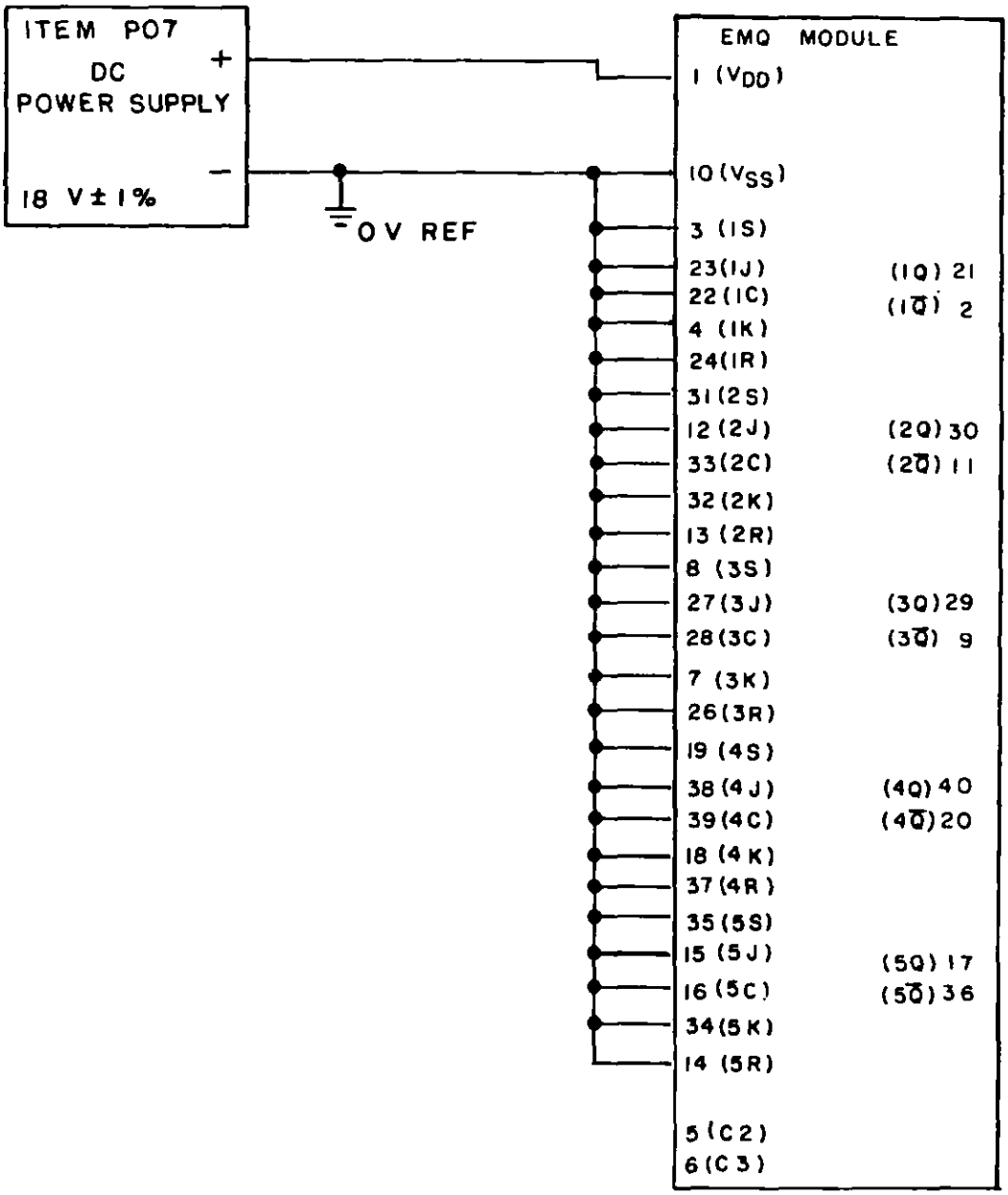


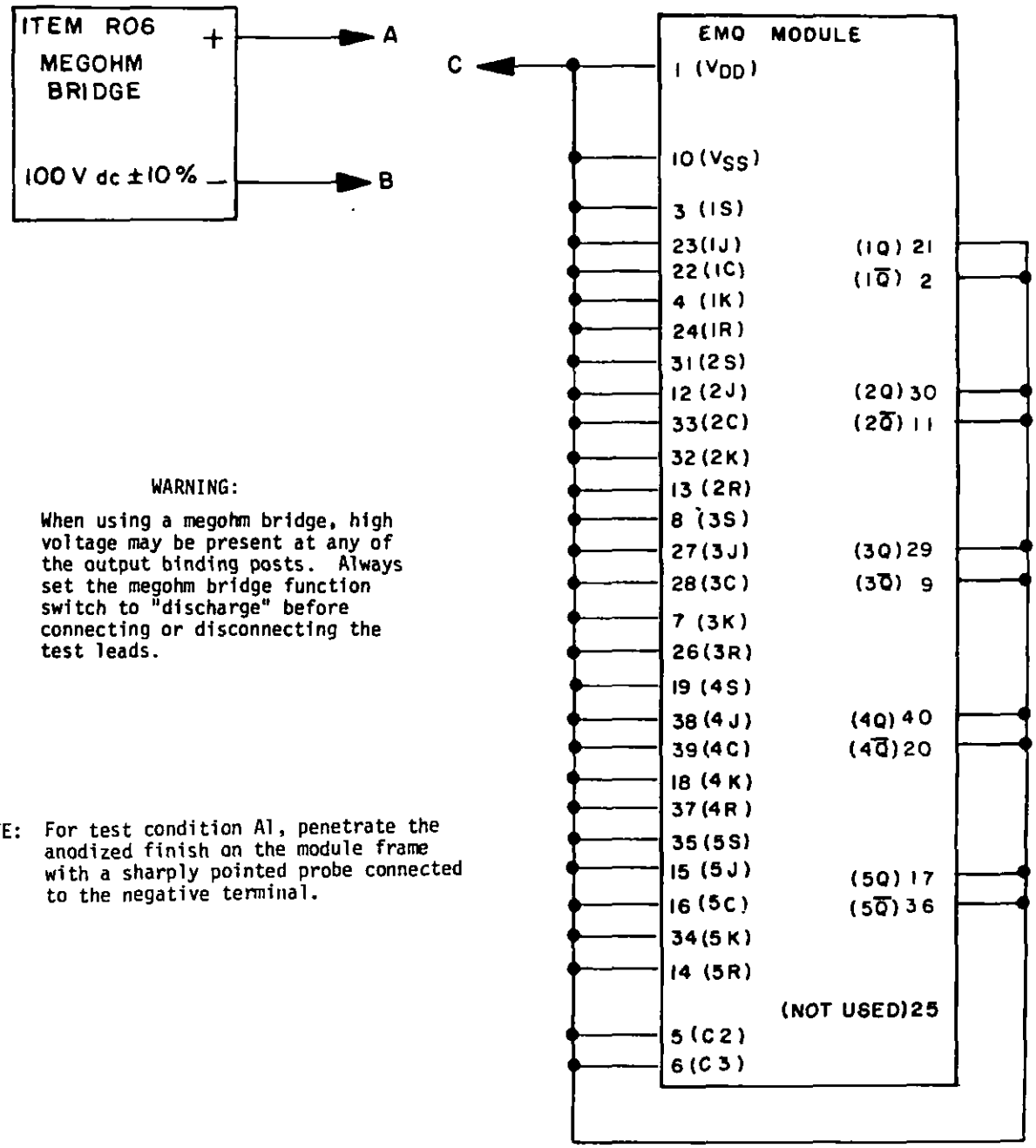
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, pin 25	Frame	10	∞	MΩ	5
A2	Isolation	pin 25	C	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Negative input clamp diode voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1S Negative input clamp diode voltage	3	Open	-0.4	-6.0	V	6
B2	1J "	23	"	"	"	"	"
B3	1C "	22	"	"	"	"	"
B4	1K "	4	"	"	"	"	"
B5	1R "	24	"	"	"	"	"
B6	2S "	31	"	"	"	"	"
B7	2J "	12	"	"	"	"	"
B8	2C "	33	"	"	"	"	"
B9	2K "	32	"	"	"	"	"
B10	2R "	13	"	"	"	"	"
B11	3S "	8	"	"	"	"	"
B12	3J "	27	"	"	"	"	"
B13	3C "	28	"	"	"	"	"
B14	3K "	7	"	"	"	"	"
B15	3R "	26	"	"	"	"	"
B16	4S "	19	"	"	"	"	"
B17	4J "	38	"	"	"	"	"
B18	4C "	39	"	"	"	"	"
B19	4K "	18	"	"	"	"	"
B20	4R "	37	"	"	"	"	"
B21	5S "	35	"	"	"	"	"

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TABLE IV. Negative input clamp diode voltage test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B22	5J Negative input clamp diode voltage	15	Open	-0.4	-6.0	V	6
B23	5C "	16	"	"	"	"	"
B24	5K "	34	"	"	"	"	"
B25	5R "	14	"	"	"	"	"

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TABLE V. Positive input clamp diode voltage test.

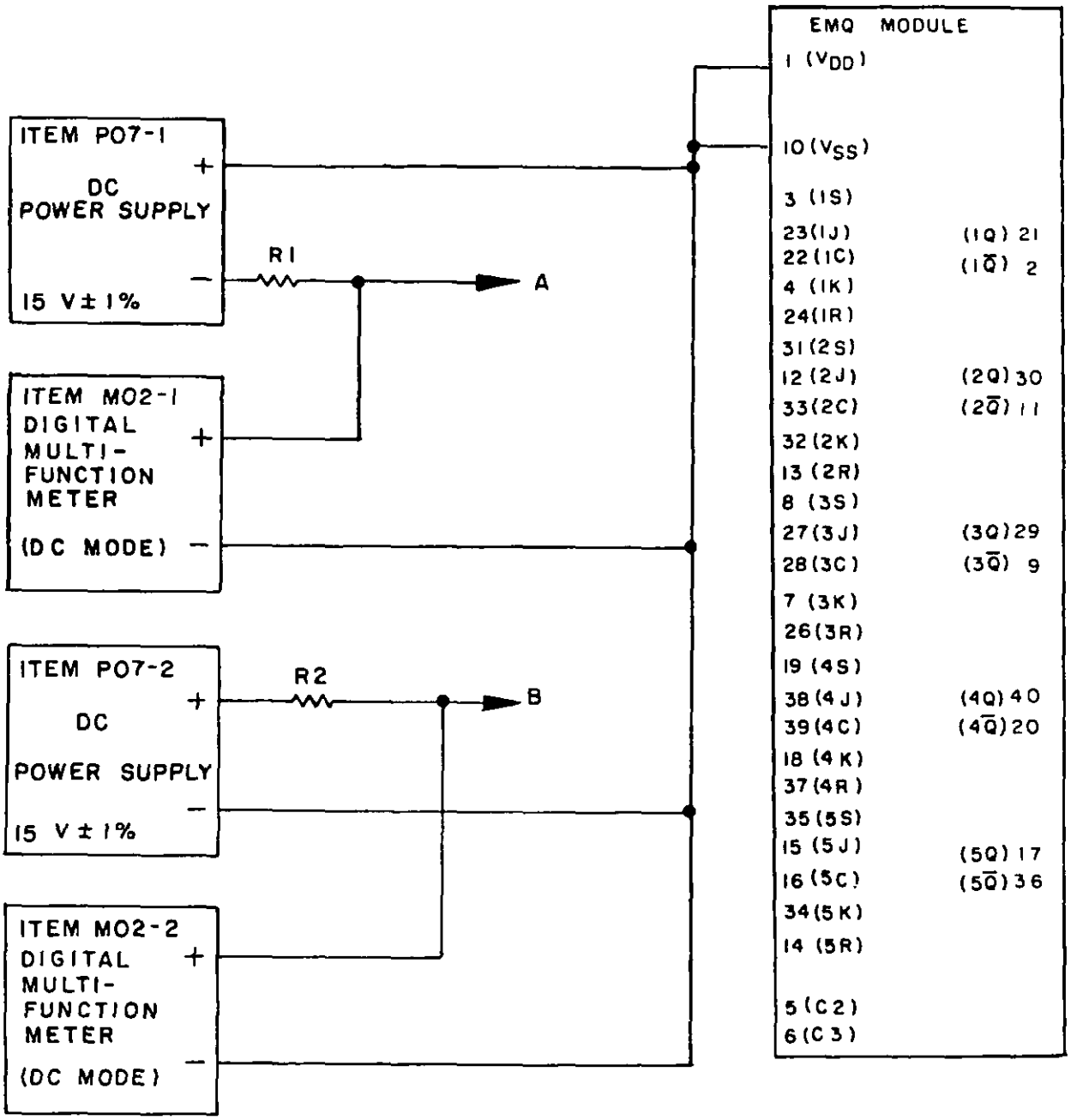
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1S Positive input clamp diode voltage	3	Open	0.4	1.5	V	6
C2	1J "	23	"	"	"	"	"
C3	1C "	22	"	"	"	"	"
C4	1K "	4	"	"	"	"	"
C5	1R "	24	"	"	"	"	"
C6	2S "	31	"	"	"	"	"
C7	2J "	12	"	"	"	"	"
C8	2C "	33	"	"	"	"	"
C9	2K "	32	"	"	"	"	"
C10	2R "	13	"	"	"	"	"
C11	3S "	8	"	"	"	"	"
C12	3J "	27	"	"	"	"	"
C13	3C "	28	"	"	"	"	"
C14	3K "	7	"	"	"	"	"
C15	3R "	26	"	"	"	"	"
C16	4S "	19	"	"	"	"	"
C17	4J "	38	"	"	"	"	"
C18	4C "	39	"	"	"	"	"
C19	4K "	18	"	"	"	"	"
C20	4R "	37	"	"	"	"	"
C21	5S "	35	"	"	"	"	"

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TABLE V. Positive input clamp diode voltage test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C22	5J Positive input clamp diode voltage	15	Open	0.4	1.5	V	6
C23	5C "	16	"	"	"	"	"
C24	5K "	34	"	"	"	"	"
C25	5R "	14	"	"	"	"	"

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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items P07-1, P07-2, M02-1, and M02-2 are not used simultaneously, only one item P07 and M02 is required.

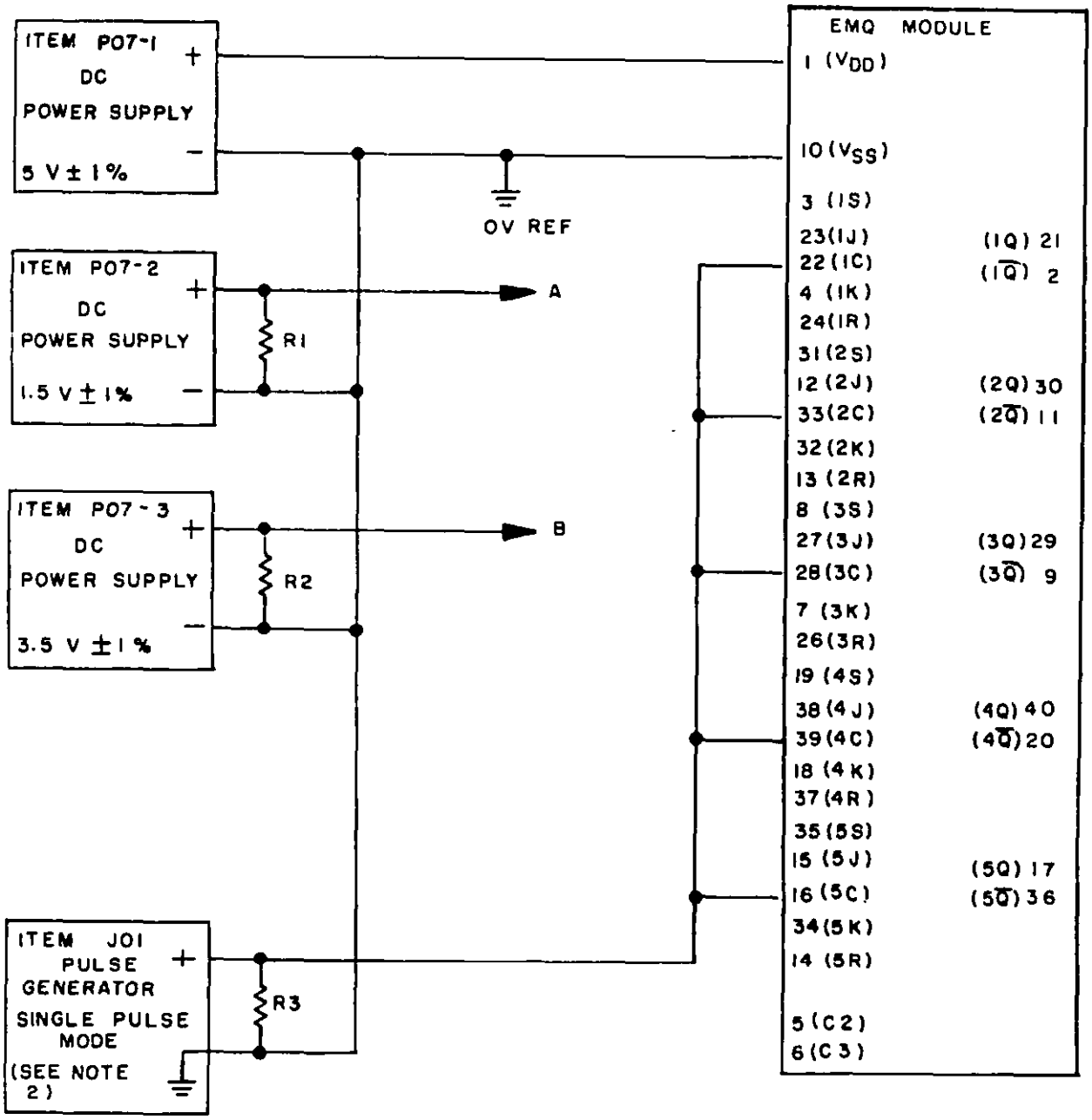
* FIGURE 6. Negative and positive input clamp diode voltage test for tables IV and V.

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TABLE VI. Functional test.

Test condition	Parameter	Connect		Output logic state										Test figure
		A to pin	B to pin	Q1	Q2	Q3	Q4	Q5	$\overline{Q1}$	$\overline{Q2}$	$\overline{Q3}$	$\overline{Q4}$	$\overline{Q5}$	
D1	Functional	4,24,32,13,7,26,18,37,34,14	3,23,31,12,8,27,19,38,35,15	H	H	H	H	H	L	L	L	L	L	7
D2	"	3,4,31,32,8,7,19,18,35,34	23,24,12,13,27,26,38,37,15,14	L	L	L	L	L	H	H	H	H	H	"
D3	"	3,4,24,31,32,13,8,7,26,19, 18,37,35,34,14	23,12,27,38,15	H	H	H	H	H	L	L	L	L	L	"
D4	"	3,23,24,31,12,13,8,27,26,19, 38,37,35,15,14	4,32,7,18,34	L	L	L	L	L	H	H	H	H	H	"
D5	"	3,24,31,13,8,26,19,37,35,14	23,4,12,32,27,7,38,18,15,34	H	H	H	H	H	L	L	L	L	L	"
D6	"	3,24,31,13,8,26,19,37,35,14	23,4,12,32,27,7,38,18,15,34	L	L	L	L	L	H	H	H	H	H	"
D7	"	3,23,4,24,31,12,32,13,8,27, 7,26,19,38,18,37,35,15,34,14	Open	L	L	L	L	L	H	H	H	H	H	"

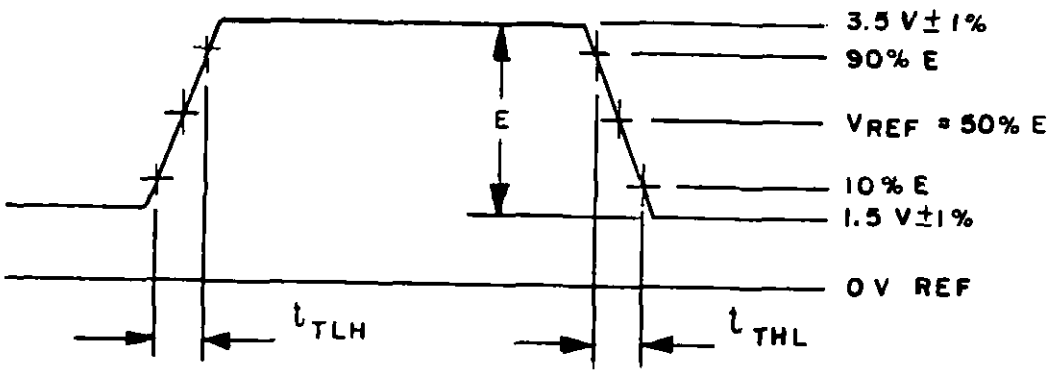
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NOTES:
1. R1 = R2 = R3 = 51.1Ω ±5%, 1/2 W.
2. After the input conditions are setup for each test condition, apply one pulse using item J01, then make the required measurement. (Test conditions D1-D7 must be done in sequence without any power interruption to V_{DD}.)

* FIGURE 7. Functional test circuit for table VI.

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3. Item J01 setup:
PW = 1 µs ±10%.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.
PRR = 500 kHz ±10%. (*The PRR is used for setup procedures only. When* the pulse has been setup, item J01 should be placed in the single cycle mode.)

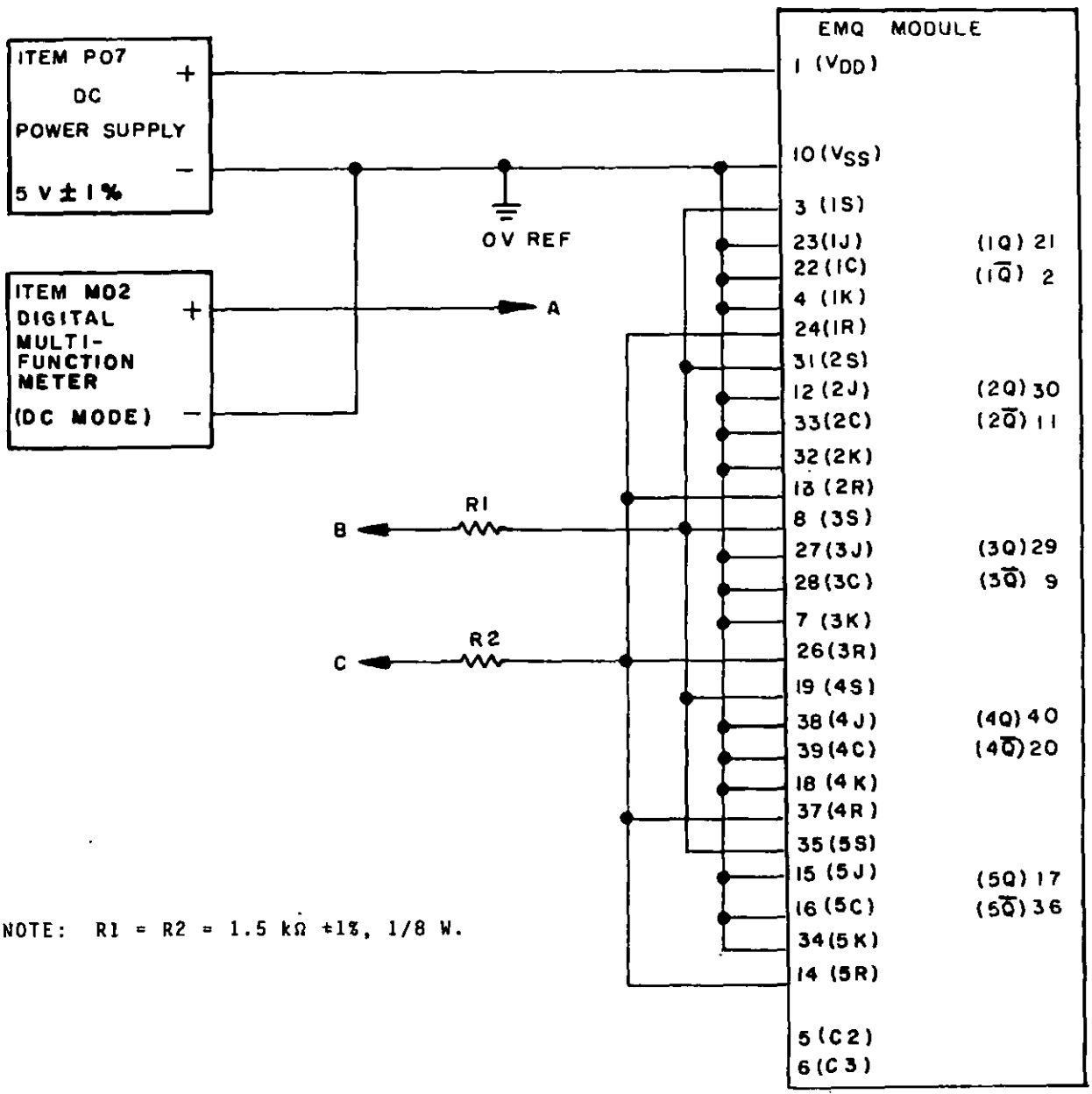
* FIGURE 7. Functional test circuit for table VI - Continued.

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*TABLE VII. Output voltage (low-level and high-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	1 Output voltage (I_{OL} and I_{OH} = 0)	Low level	21	10	1	0	50	mV	8
F1	1 "	High level	2	"	"	4.95	5.05	V	"
E2	2 "	Low level	30	"	"	0	50	mV	"
F2	2 "	High level	11	"	"	4.95	5.05	V	"
E3	3 "	Low level	29	"	"	0	50	mV	"
F3	3 "	High level	9	"	"	4.95	5.05	V	"
E4	4 "	Low level	40	"	"	0	50	mV	"
F4	4 "	High level	20	"	"	4.95	5.05	V	"
E5	5 "	Low level	17	"	"	0	50	mV	"
F5	5 "	High level	36	"	"	4.95	5.05	V	"
F6	6 "	High level	21	1	10	4.95	5.05	V	"
E6	6 "	Low level	2	"	"	0	50	mV	"
F7	7 "	High level	30	"	"	4.95	5.05	V	"
E7	7 "	Low level	11	"	"	0	50	mV	"
F8	8 "	High level	29	"	"	4.95	5.05	V	"
E8	8 "	Low level	9	"	"	0	50	mV	"
F9	9 "	High level	40	"	"	4.95	5.05	V	"
E9	9 "	Low level	20	"	"	0	50	mV	"
F10	10 "	High level	17	"	"	4.95	5.05	V	"
E10	10 "	Low level	36	"	"	0	50	mV	"

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NOTE: R1 = R2 = 1.5 kΩ ±1%, 1/8 W.

* FIGURE 8. Output voltage (low-level and high-level) test circuit for table VII.

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*TABLE VIII. Output voltage (low-level) test.

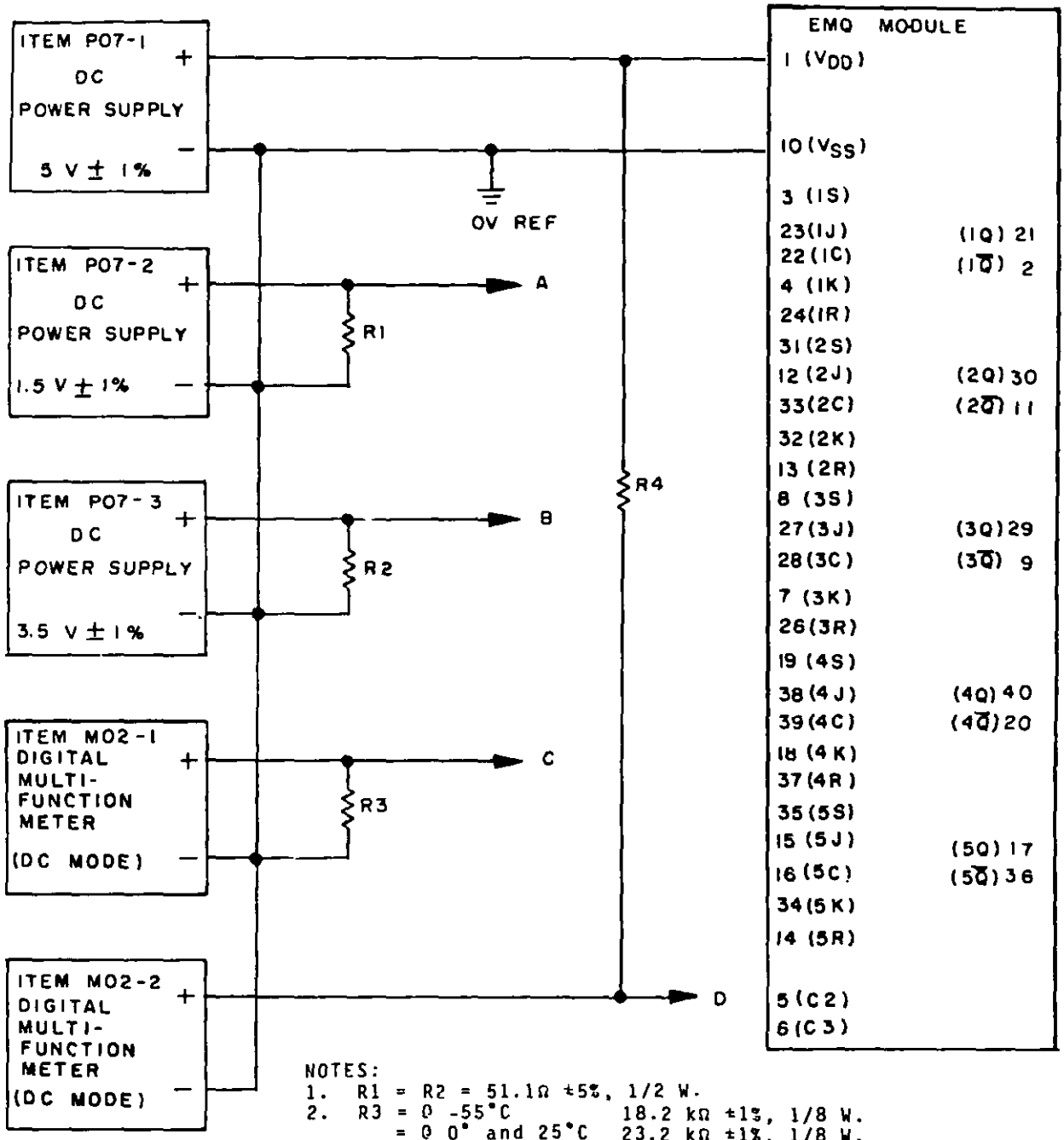
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1Q Output voltage, low level (V_{OD} = 5 V)	3,23,22,4,12,33,32,13, 27,28,7,26,38,39,18, 37,15,16,34,14	24,31,8,19,35	Open	21	0	400	mV	9
G2	2Q "	23,22,4,24,31,12,33, 32,27,28,7,26,38,39, 18,37,15,16,34,14	3,13,8,19,35	"	30	"	"	"	"
G3	3Q "	23,22,4,24,12,33,32, 13,8,27,28,7,38,39,18, 37,15,16,34,14	3,31,26,19,35	"	29	"	"	"	"
G4	4Q "	23,22,4,24,12,33,32, 13,27,28,7,26,19,38, 39,18,15,16,34,14	3,31,8,37,35	"	40	"	"	"	"
G5	5Q "	23,22,4,24,12,33,32, 13,27,28,7,26,38,39, 18,37,35,15,16,34	3,31,8,19,14	"	17	"	"	"	"
G6	$1\overline{Q}$ "	24,31,8,19,35	3,23,22,4,12,33,32, 13,27,28,7,26,38,39, 18,37,15,16,34,14	"	2	"	"	"	"
G7	$2\overline{Q}$ "	3,13,8,19,35	23,22,4,24,31,12,33, 32,27,28,7,26,38,39, 18,37,15,16,34,14	"	11	"	"	"	"
G8	$3\overline{Q}$ "	3,31,26,19,35	23,22,4,24,12,33,32, 13,8,27,28,7,38,39, 18,37,15,16,34,14	"	9	"	"	"	"
G9	$4\overline{Q}$ "	3,31,8,37,35	23,22,4,24,12,33,32, 13,27,28,7,26,19,38, 39,18,15,16,34,14	"	20	"	"	"	"
G10	$5\overline{Q}$ "	3,31,8,19,14	23,22,4,24,12,33,32, 13,27,28,7,26,38,39, 18,37,35,15,16,34	"	36	"	"	"	"

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*TABLE IX. Output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1Q Output voltage, high level (V_{DD} = 5 V)	24,31,8,19,35	3,23,22,4,12,33,32,13,27,28,7,26,38,39,18,37,15,16,34,14	21	Open	4.6	5.0	V	9
H2	2Q "	3,13,8,19,35	23,22,4,24,31,12,33,32,27,28,7,26,38,39,18,37,15,16,34,14	30	"	"	"	"	"
H3	3Q "	3,31,26,19,35	23,22,4,24,12,33,32,13,8,27,28,7,38,39,18,37,15,16,34,14	29	"	"	"	"	"
H4	4Q "	3,31,8,37,35	23,22,4,24,12,33,32,13,27,28,7,26,19,38,39,18,15,16,34,14	40	"	"	"	"	"
H5	5Q "	3,31,8,19,14	23,22,4,24,12,33,32,13,27,28,7,26,38,39,18,37,35,15,16,34	17	"	"	"	"	"
H6	$1\overline{Q}$ "	3,23,22,4,12,33,32,13,27,28,7,26,38,39,18,37,15,16,34,14	24,31,8,19,35	2	"	"	"	"	"
H7	$2\overline{Q}$ "	23,22,4,24,31,12,33,32,27,14,28,7,26,38,39,18,37,15,16,34	3,13,8,19,35	11	"	"	"	"	"
H8	$3\overline{Q}$ "	23,22,4,24,12,33,32,13,8,27,28,7,38,39,18,37,15,16,34,14	3,31,26,19,35	9	"	"	"	"	"
H9	$4\overline{Q}$ "	23,22,4,24,12,33,32,13,27,28,7,26,19,38,39,18,15,16,34,14	3,31,8,37,35	20	"	"	"	"	"
H10	$5\overline{Q}$ "	23,22,4,24,12,33,32,13,27,28,7,26,38,39,18,37,35,15,16,34	3,31,8,19,14	36	"	"	"	"	"

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 0 -55°C 18.2 kΩ ±1%, 1/8 W.
   = 0 0° and 25°C 23.2 kΩ ±1%, 1/8 W.
   = 0 60°C 27.4 kΩ ±1%, 1/8 W.
   = 0 85°C 32.4 kΩ ±1%, 1/8 W.
3. R4 = 0 -55°C 7.15 kΩ ±1%, 1/8 W.
   = 0 0° and 25°C 9.09 kΩ ±1%, 1/8 W.
   = 0 60°C 10.5 kΩ ±1%, 1/8 W.
   = 0 85°C 12.7 kΩ ±1%, 1/8 W.

* FIGURE 9. Output voltage (high-level and low-level) test circuit for tables VIII and IX.

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*TABLE X. Output voltage (low-level) test.

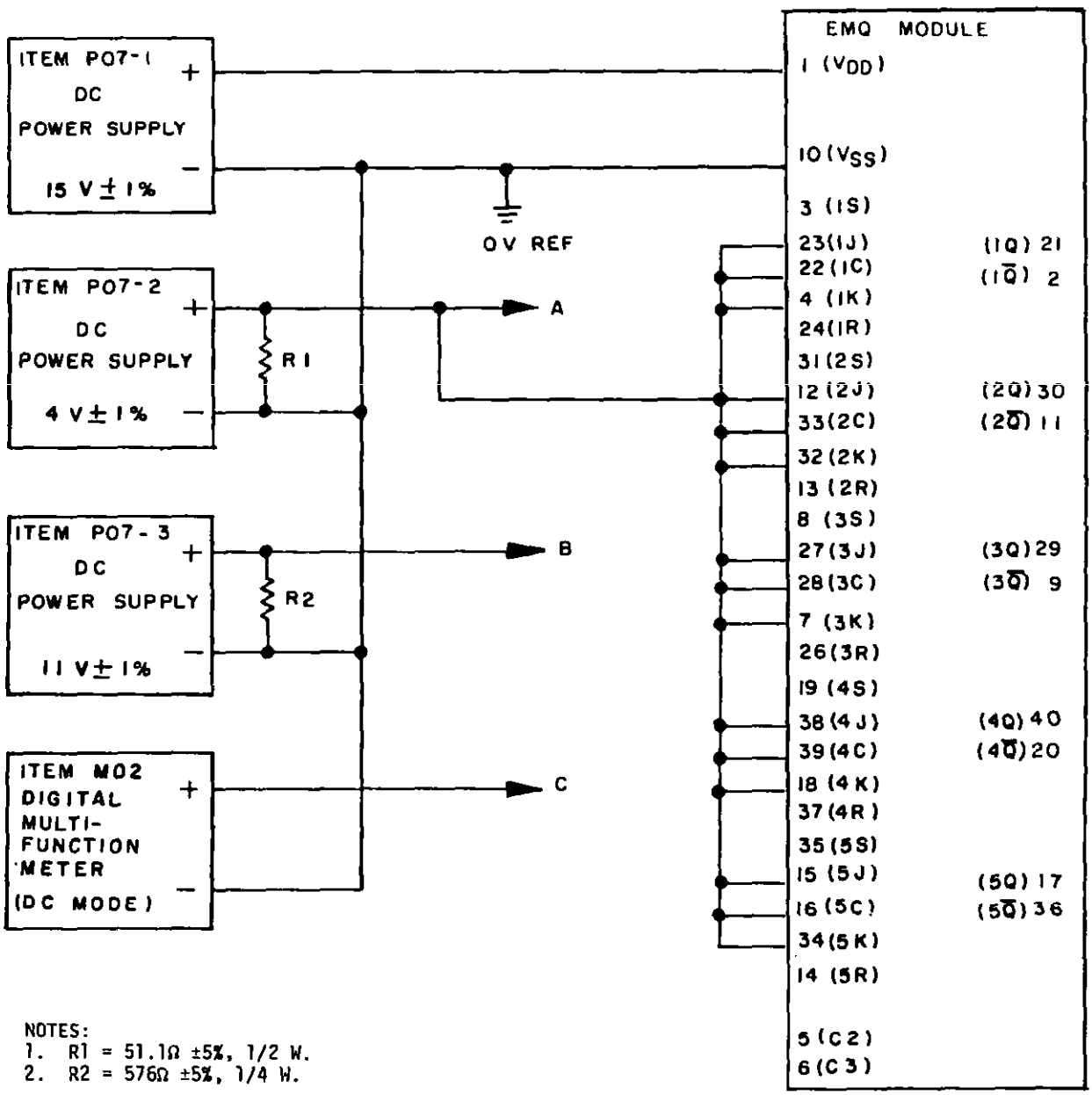
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1Q Output voltage, low level (V_{DD} = 15 V)	3,31,8, 19,35	24,13,26, 37,14	21	0.0	1.5	V	10
J2	2Q "	"	"	30	"	"	"	"
J3	3Q "	"	"	29	"	"	"	"
J4	4Q "	"	"	40	"	"	"	"
J5	5Q "	"	"	17	"	"	"	"
J6	$1\overline{Q}$ "	24,13,26, 37,14	3,31,8, 19,35	2	"	"	"	"
J7	$2\overline{Q}$ "	"	"	11	"	"	"	"
J8	$3\overline{Q}$ "	"	"	9	"	"	"	"
J9	$4\overline{Q}$ "	"	"	20	"	"	"	"
J10	$5\overline{Q}$ "	"	"	36	"	"	"	"

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*TABLE XI. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1Q Output voltage, high level (V_{DD} = 15 V)	24,13,26, 37,14	3,31,8, 19,35,	21	13.5	15.0	V	10
K2	2Q "	"	"	30	"	"	"	"
K3	3Q "	"	"	29	"	"	"	"
K4	4Q "	"	"	40	"	"	"	"
K5	5Q "	"	"	17	"	"	"	"
K6	$1\overline{Q}$ "	3,31,8, 19,35	24,13,26, 37,14	2	"	"	"	"
K7	$2\overline{Q}$ "	"	"	11	"	"	"	"
K8	$3\overline{Q}$ "	"	"	9	"	"	"	"
K9	$4\overline{Q}$ "	"	"	20	"	"	"	"
K10	$5\overline{Q}$ "	"	"	36	"	"	"	"

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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 576Ω ±5%, 1/4 W.

* FIGURE 10. Output voltage (low-level and high-level) test circuit for tables X and XI.

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TABLE XII. Input current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1S Input current, low level	23,22,4,24,31,12,33,32,13,8, 27,28,7,26,19,38,39,18,37,35, 15,16,34,14	3	0	-200	nA	11
L2	1J "	3,22,4,24,31,12,33,32,13,8, 27,28,7,26,19,38,39,18,37,35, 15,16,34,14	23	"	"	"	"
L3	1C "	3,23,4,24,31,12,33,32,13,8, 27,28,26,19,7,38,39,18,37,35, 15,16,34,14	22	"	"	"	"
L4	1K "	3,23,22,24,31,12,33,32,13,8, 27,28,7,26,19,38,39,18,37,35, 15,16,34,14	4	"	"	"	"
L5	1R "	3,23,22,4,31,12,33,32,13,8, 27,28,7,26,19,38,39,18,37,35, 15,16,34,14	24	"	"	"	"
L6	2S "	3,23,22,4,24,12,33,32,13,8, 27,28,26,19,7,38,39,18,37,35, 15,16,34,14	31	"	"	"	"
L7	2J "	3,23,22,4,24,31,33,32,13,8, 27,28,7,26,19,38,39,18,37,35, 15,16,34,14	12	"	"	"	"
L8	2C "	3,23,22,4,24,31,12,32,13,8, 27,28,7,26,19,38,39,18,37,35, 15,16,34,14	33	"	"	"	"
L9	2K "	3,23,22,4,24,31,12,33,13,8, 27,28,7,26,19,38,39,18,37,35, 15,16,34,14	32	"	"	"	"
L10	2R "	3,23,22,4,24,31,12,33,32,8, 27,28,7,26,19,38,39,18,37,35, 15,16,34,14	13	"	"	"	"
L11	3S "	3,23,22,4,24,31,12,33,32,13, 27,28,7,26,19,38,39,18,37,35, 15,16,34,14	8	"	"	"	"

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TABLE XII. Input current (low-level) test - Continued.

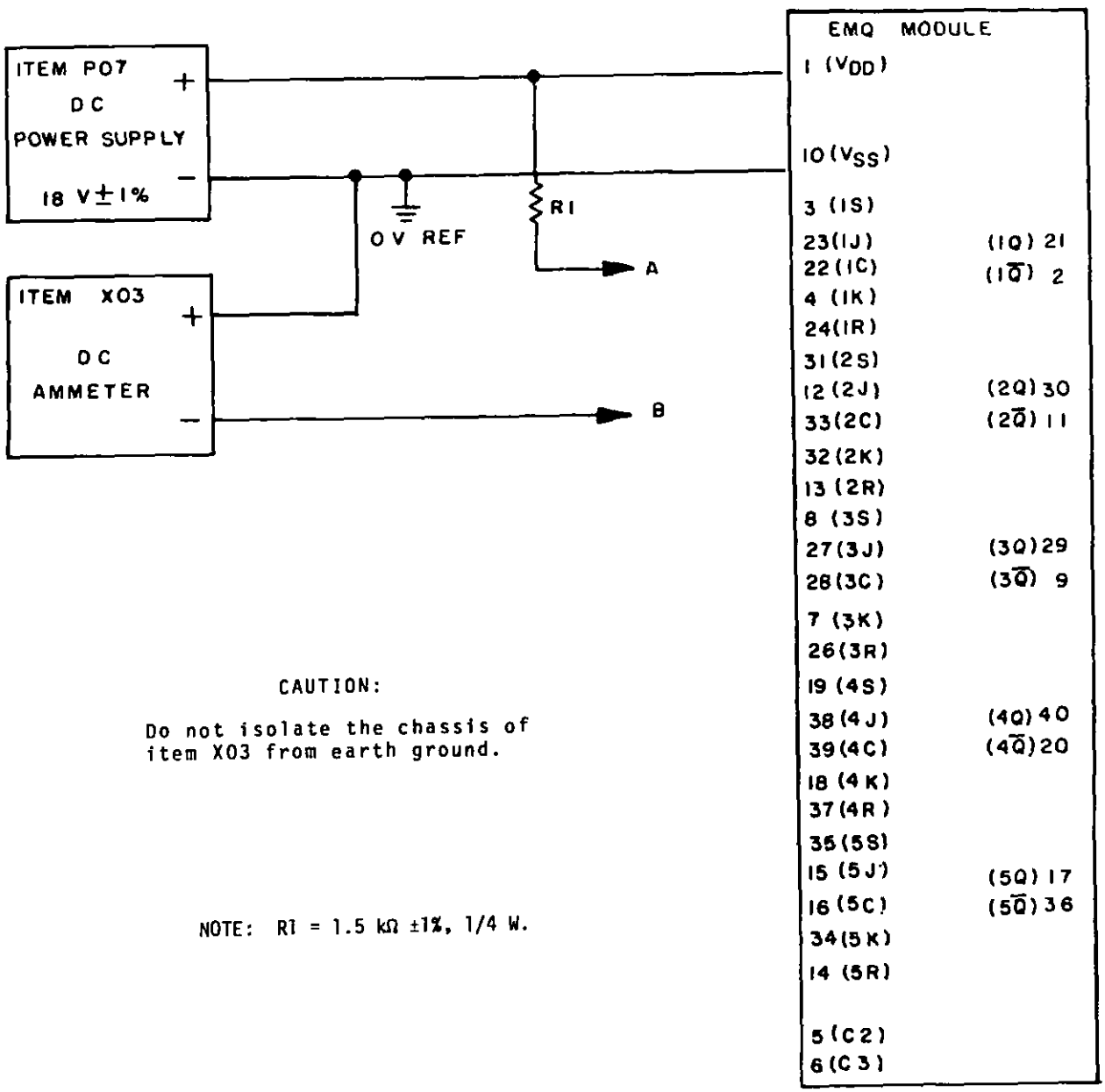
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L12	3J Input current, low level	3,23,22,4,24,31,12,33,32,13, 8,28,7,26,19,38,39,18,37,35, 15,16,34,14	27	0	-200	nA	11
L13	3C "	3,23,22,4,24,31,12,33,32,13, 8,27,7,26,19,38,39,18,37,35, 15,16,34,14	28	"	"	"	"
L14	3K "	3,23,22,4,24,31,12,33,32,13, 8,27,28,26,19,38,39,18,37,35, 15,16,34,14	7	"	"	"	"
L15	3R "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,19,38,39,18,37,35, 15,16,34,14	26	"	"	"	"
L16	4S "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,38,39,18,37,35, 15,16,34,14	19	"	"	"	"
L17	4J "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,39,18,37,35, 15,16,34,14	38	"	"	"	"
L18	4C "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,18,37,35, 15,16,34,14	39	"	"	"	"
L19	4K "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,37,35, 15,16,34,14	18	"	"	"	"
L20	4R "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,35, 15,16,34,14	37	"	"	"	"
L21	5S "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,37, 15,16,34,14	35	"	"	"	"
L22	5J "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,37, 35,16,34,14	15	"	"	"	"

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L23	5C Input current, low level	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,37, 35,15,34,14	16	0	-200	nA	11
L24	5K "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,37, 35,15,16,14	34	"	"	"	"
L25	5R "	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,37, 35,15,16,34	14	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO3 from earth ground.

NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 11. Input current (low-level) test circuit for table XII.

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TABLE XIII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	1S Input current, high level	3	23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	0	200	nA	12
M2	1J "	23	3,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	"	"	"	"
M3	1C "	22	3,23,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,16,34,14,15	"	"	"	"
M4	1K "	4	3,23,22,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	"	"	"	"
M5	1R "	24	3,23,22,4,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	"	"	"	"
M6	2S "	31	3,23,22,4,24,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	"	"	"	"
M7	2J "	12	3,23,22,4,24,31,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	"	"	"	"
M8	2C "	33	3,23,22,4,24,31,12,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	"	"	"	"
M9	2K "	32	3,23,22,4,24,31,12,33,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	"	"	"	"
M10	2R "	13	3,23,22,4,24,31,12,33,32, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	"	"	"	"
M11	3S "	8	3,23,22,4,24,31,12,33,32, 13,28,27,7,26,19,38,39,18, 37,35,15,16,34,14	"	"	"	"

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Check the source to verify that this is the current version before use.

TABLE XIII. Input current (high-level) test - Continued.

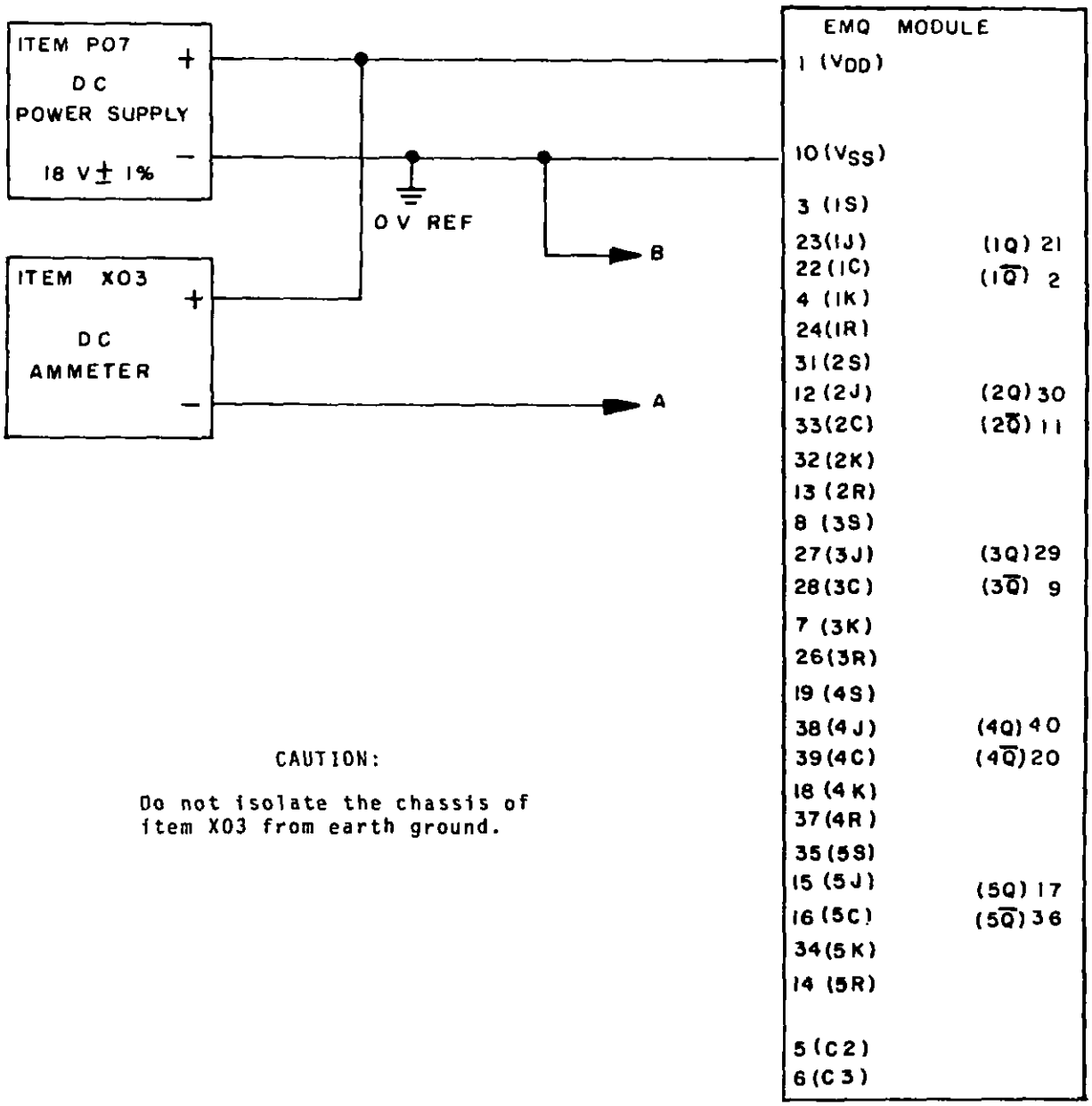
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M12	3J Input current, high level	27	3,23,22,4,24,31,12,33,32,13,8,28,7,26,19,38,39,18,37,35,15,16,34,14	0	200	nA	12
M13	3C "	28	3,23,22,4,24,31,12,33,32,13,8,27,7,26,19,38,39,18,37,35,15,16,34,14	"	"	"	"
M14	3K "	7	3,23,22,4,24,31,12,33,32,13,8,27,28,26,19,38,39,18,37,35,15,16,34,14	"	"	"	"
M15	3R "	26	3,23,22,4,24,31,12,33,32,13,8,27,28,7,19,38,39,18,37,35,15,16,34,14	"	"	"	"
M16	4S "	19	3,23,22,4,24,31,12,33,32,13,8,27,28,7,26,38,39,18,37,35,15,16,34,14	"	"	"	"
M17	4J "	38	3,23,22,4,24,31,12,33,32,13,8,27,28,7,26,19,39,18,37,35,15,16,34,14	"	"	"	"
M18	4C "	39	3,23,22,4,24,31,12,33,32,13,8,27,28,7,26,19,38,18,37,35,15,16,34,14	"	"	"	"
M19	4K "	18	3,23,22,4,24,31,12,33,32,13,8,27,28,7,26,19,38,39,37,35,15,16,34,14	"	"	"	"
M20	4R "	37	3,23,22,4,24,31,12,33,32,13,8,27,28,7,26,19,38,39,18,35,15,16,34,14	"	"	"	"
M21	5S "	35	3,23,22,4,24,31,12,33,32,13,8,27,28,7,26,19,38,39,18,37,15,16,34,14	"	"	"	"
M22	5J "	15	3,23,22,4,24,31,12,33,32,13,8,27,28,7,26,19,38,39,18,37,35,16,34,14	"	"	"	"

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M23	5C Input current, high level	16	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 18,37,35,15,34,14	0	200	nA	12
M24	5K "	34	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 18,37,35,15,16,14	"	"	"	"
M25	5R "	14	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 18,37,35,15,16,34	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X03 from earth ground.

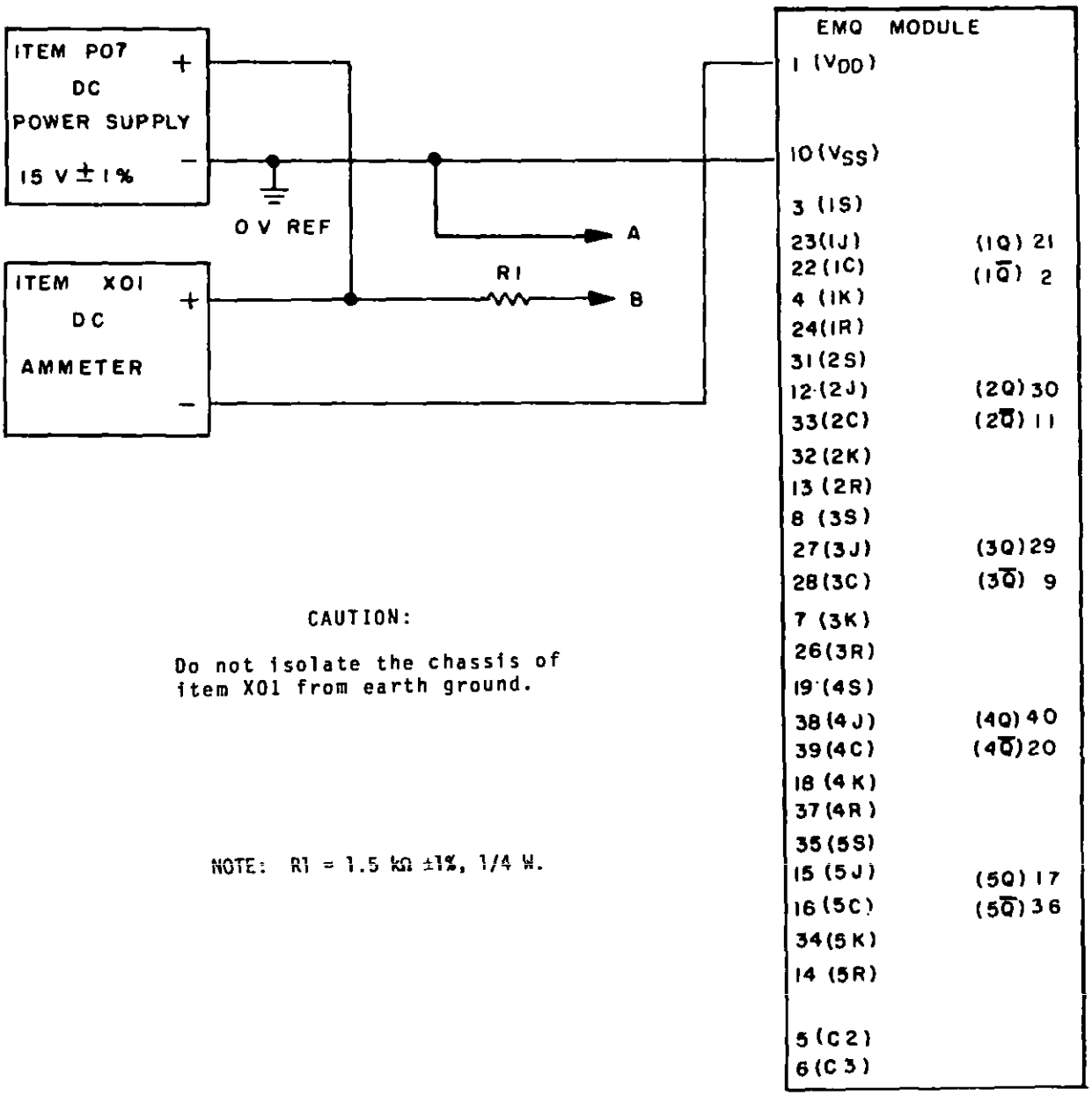
FIGURE 12. Input current (high-level) test circuit for table XIII.

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*TABLE XIV. Power supply current (low and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Power supply current, low level	3,31,8,19,35	23,22,4,24,12,33,32,13,27, 28,7,26,38,39,18,37,15,16, 34,14	0	60	µA	13
N2	Power supply current, high level	23,22,4,24,12,33,32,13,27, 28,7,26,38,39,18,37,15,16, 34,14	3,31,8,19,35	0	60	µA	13

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

* FIGURE 13. Power supply current (low-level and high-level) test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1Q Transition time, C_L = 50 pF, V_{DD} = 5V	t_{TLH}	12,33,27,28,38, 39,15,16	23	22	21	36	360	ns	14,15
Q1	$1\overline{Q}$ "	t_{THL}	12,33,27,28,38, 39,15,16	23	22	2	20	200	"	"
P2	2Q "	t_{TLH}	23,22,27,28,38, 39,15,16	12	33	30	36	360	"	"
Q2	$2\overline{Q}$ "	t_{THL}	23,22,27,28,38, 39,15,16	12	33	11	20	200	"	"
P3	3Q "	t_{TLH}	23,22,12,33,38, 39,15,16	27	28	29	36	360	"	"
Q3	$3\overline{Q}$ "	t_{THL}	23,22,12,33,38, 39,15,16	27	28	9	20	200	"	"
P4	4Q "	t_{TLH}	23,22,12,33,27, 28,15,16	38	39	40	36	360	"	"
Q4	$4\overline{Q}$ "	t_{THL}	23,22,12,33,27, 28,15,16	38	39	20	20	200	"	"
P5	5Q "	t_{TLH}	23,22,12,33,27, 28,38,39	15	16	17	36	360	"	"
Q5	$5\overline{Q}$ "	t_{THL}	23,22,12,33,27, 28,38,39	15	16	36	20	200	"	"
Q6	1Q "	t_{THL}	12,33,27,28,38, 39,15,16	23	22	21	20	200	"	"
P6	$1\overline{Q}$ "	t_{TLH}	12,33,27,28,38, 39,15,16	23	22	2	36	360	"	"
Q7	2Q "	t_{THL}	23,22,27,28,38, 39,15,16	12	33	30	20	200	"	"
P7	$2\overline{Q}$ "	t_{TLH}	23,22,27,28,38, 39,15,16	12	33	11	36	360	"	"

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Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q8	3Q Transition time, C_L = 50 pF, V_{DD} = 5 V	t_{THL}	23,22,12,33,38, 39,15,16	27	28	29	20	200	ns	14,15
P8	$3\overline{Q}$ "	t_{TLH}	23,22,12,33,38, 39,15,16	27	28	9	36	360	"	"
Q9	4Q "	t_{THL}	23,22,12,33,27, 28,15,16	38	39	40	20	200	"	"
P9	$4\overline{Q}$ "	t_{TLH}	23,22,12,33,27, 28,15,16	38	39	20	36	360	"	"
Q10	5Q "	t_{THL}	23,22,12,33,27, 28,38,39	15	16	17	20	200	"	"
P10	$5\overline{Q}$ "	t_{TLH}	23,22,12,33,27, 28,38,39	15	16	36	36	360	"	"
P11	1Q Transition time, C_L = 150 pF, V_{DD} = 5V	t_{TLH}	12,33,27,28,38, 39,15,16	23	22	21	84	840	"	"
Q11	$1\overline{Q}$ "	t_{THL}	12,33,27,28,38, 39,15,16	23	22	2	68	680	"	"
P12	2Q "	t_{TLH}	23,22,27,28,38, 39,15,16	12	33	30	84	840	"	"
Q12	$2\overline{Q}$ "	t_{THL}	23,22,27,28,38, 39,15,16	12	33	11	68	680	"	"
P13	3Q "	t_{TLH}	23,22,12,33,38, 39,15,16	27	28	29	84	840	"	"
Q13	$3\overline{Q}$ "	t_{THL}	23,22,12,13,38, 39,15,16	27	28	9	68	680	"	"
P14	4Q "	t_{TLH}	23,22,12,33,27, 28,15,16	38	39	40	84	840	"	"
Q14	$4\overline{Q}$ "	t_{THL}	23,22,12,33,27, 28,15,16	38	39	20	68	680	"	"

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T17:13Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P15	5Q Transition time, C_L = 150 pF, V_{DD} = 5 V	t_{TLH}	23,22,17,33,27, 28,38,39	15	16	17	84	840	ns	14,15
Q15	$5\overline{Q}$ "	t_{THL}	23,22,17,33,27, 28,38,39	15	16	36	68	680	"	"
Q16	1Q "	t_{THL}	12,33,27,28,38, 39,15,16	23	22	21	68	680	"	"
P16	$1\overline{Q}$ "	t_{TLH}	12,33,27,28,38, 39,15,16	23	22	2	84	840	"	"
Q17	2Q "	t_{THL}	23,22,27,28,38, 39,15,16	12	33	30	68	680	"	"
P17	$2\overline{Q}$ "	t_{TLH}	23,22,27,28,38, 39,15,16	12	33	11	84	840	"	"
Q18	3Q "	t_{THL}	23,22,12,33,38, 39,15,16	27	28	29	68	680	"	"
P18	$3\overline{Q}$ "	t_{TLH}	23,22,12,33,38, 39,15,16	27	28	9	84	840	"	"
Q19	4Q "	t_{THL}	23,22,12,33,27, 28,15,16	38	39	40	68	680	"	"
P19	$4\overline{Q}$ "	t_{TLH}	23,22,12,33,27, 28,15,16	38	39	20	84	840	"	"
Q20	5Q "	t_{THL}	23,22,12,33,27, 28,38,39	15	16	17	68	680	"	"
P20	$5\overline{Q}$ "	t_{TLH}	23,22,12,33,27, 28,38,39	15	16	36	84	840	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1Q Propagation delay time, C_L = 50 pF, (clock to output), V_{DD} = 5 V	t_{PLH}	12,23,27,28,38, 39,15,16	23	22	21	35	350	ns	14,16
S1	$1\overline{Q}$	" t_{PHL}	12,33,27,28,38, 39,15,16	23	22	2	"	"	"	"
R2	2Q	" t_{PLH}	23,22,27,28,38, 39,15,16	12	33	30	"	"	"	"
S2	$2\overline{Q}$	" t_{PHL}	23,22,27,28,38, 39,15,16	12	33	11	"	"	"	"
R3	3Q	" t_{PLH}	23,22,12,33,38, 39,15,16	27	28	29	"	"	"	"
S3	$3\overline{Q}$	" t_{PHL}	23,22,12,33,38, 39,15,16	27	28	9	"	"	"	"
R4	4Q	" t_{PLH}	23,22,12,33,27, 28,15,16	38	39	40	"	"	"	"
S4	$4\overline{Q}$	" t_{PHL}	23,22,12,33,27, 28,15,16	38	39	20	"	"	"	"
R5	5Q	" t_{PLH}	23,22,12,33,27, 28,38,39	15	16	17	"	"	"	"
S5	$5\overline{Q}$	" t_{PHL}	23,22,12,33,27, 28,38,39	15	16	36	"	"	"	"
S6	1Q	" t_{PHL}	12,33,27,28,38, 39,15,16	23	22	21	"	"	"	"
R6	$1\overline{Q}$	" t_{PLH}	12,33,27,28,38, 39,15,16	23	22	2	"	"	"	"
S7	2Q	" t_{PHL}	23,22,27,28,38, 39,15,16	12	33	30	"	"	"	"
R7	$2\overline{Q}$	" t_{PLH}	23,22,27,28,38, 39,15,16	12	33	11	"	"	"	"

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TABLE XVI. *Propagation delay time (t_{PLH} and t_{PHL}) test* - Continued.

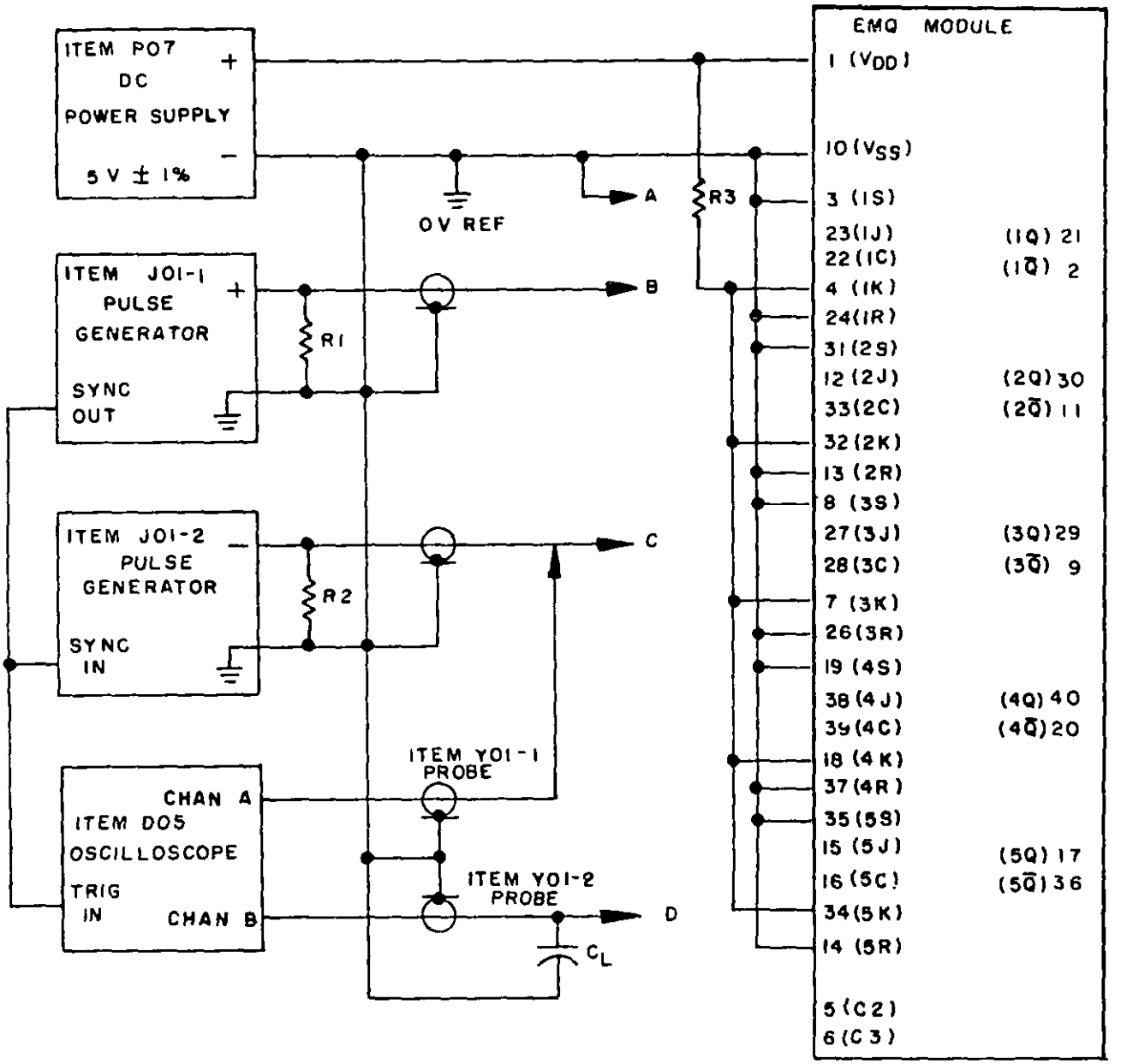
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S8	3Q Propagation delay time, C_L = 50 pF, (clock to output), V_{DD} = 5 V	t_{PHL}	23,22,27,28,38, 39,15,16	27	28	29	35	350	ns	14,16
R8	$3\overline{Q}$ "	t_{PLH}	23,22,12,33,38, 39,15,16	27	28	9	"	"	"	"
S9	4Q "	t_{PHL}	23,22,12,33,27, 28,15,16	38	39	40	"	"	"	"
R9	$4\overline{Q}$ "	t_{PLH}	23,22,12,33,27, 28,15,16	38	39	20	"	"	"	"
S10	5Q "	t_{PHL}	23,22,12,33,27, 28,38,39	15	16	17	"	"	"	"
R10	$5\overline{Q}$ "	t_{PLH}	23,22,12,33,27, 28,38,39	15	16	36	"	"	"	"
R11	1Q Propagation delay time, C_L = 150 pF, (clock to output), V_{DD} = 5 V	t_{PLH}	12,33,27,28,38, 39,15,16	23	22	21	55	550	"	"
S11	$1\overline{Q}$ "	t_{PHL}	12,33,27,28,38, 39,15,16	23	22	2	"	"	"	"
R12	2Q "	t_{PLH}	23,22,27,28,38, 39,15,16	12	33	30	"	"	"	"
S12	$2\overline{Q}$ "	t_{PHL}	23,22,27,28,38, 39,15,16	12	33	11	"	"	"	"
R13	3Q "	t_{PLH}	23,22,12,33,38, 39,15,16	27	28	29	"	"	"	"
S13	$3\overline{Q}$ "	t_{PHL}	23,22,12,33,38, 39,15,16	27	28	9	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R14	4Q Propagation delay time, C_L = 150 pF (clock to output), V_{DD} = 5 V	t_{PLH}	23,22,12,33,27, 28,15,16	38	39	40	55	550	ns	14,16
S14	$4\overline{Q}$ "	t_{PHL}	23,22,12,33,27, 28,15,16	38	39	20	"	"	"	"
R15	5Q "	t_{PLH}	23,22,12,33,27, 28,38,39	15	16	17	"	"	"	"
S15	$5\overline{Q}$ "	t_{PHL}	23,22,12,33,27, 28,38,39	15	16	36	"	"	"	"
S16	1Q "	t_{PHL}	12,33,27,28,38, 39,15,16	23	22	21	"	"	"	"
R16	$1\overline{Q}$ "	t_{PLH}	12,33,27,28,38, 39,15,16	23	22	2	"	"	"	"
S17	2Q "	t_{PHL}	23,22,27,28,38, 39,15,16	12	33	30	"	"	"	"
R17	$2\overline{Q}$ "	t_{PLH}	23,22,27,28,38, 39,15,16	12	33	11	"	"	"	"
S18	3Q "	t_{PHL}	23,22,12,33,38, 39,15,16	27	28	29	"	"	"	"
R18	$3\overline{Q}$ "	t_{PLH}	23,22,12,33,38, 39,15,16	27	28	9	"	"	"	"
S19	4Q "	t_{PHL}	23,22,12,33,27, 28,15,16	38	39	40	"	"	"	"
R19	$4\overline{Q}$ "	t_{PLH}	23,22,12,33,27, 28,15,16	38	39	20	"	"	"	"
S20	5Q "	t_{PHL}	23,22,12,33,27, 28,38,39	15	16	17	"	"	"	"
R20	$5\overline{Q}$ "	t_{PLH}	23,22,12,33,27, 28,38,39	15	16	36	"	"	"	"

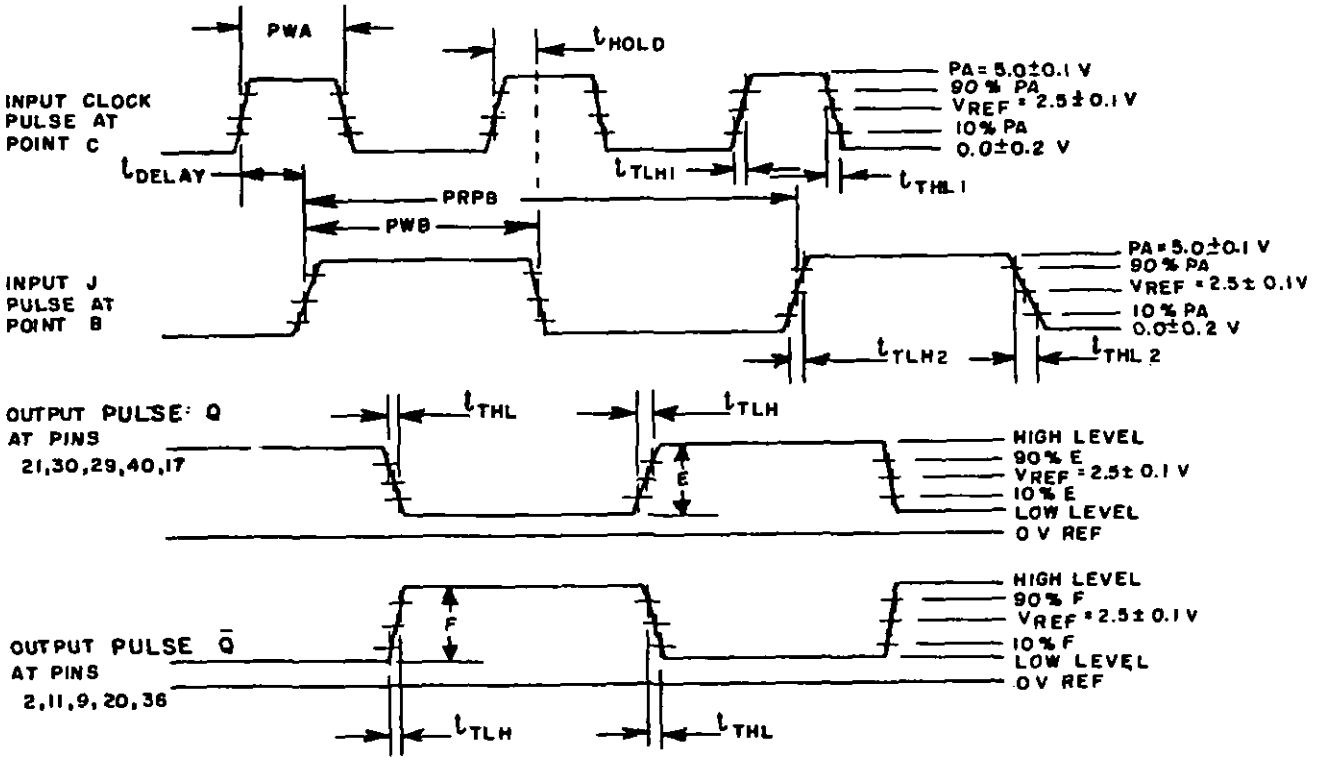
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions P1-P10, Q1-Q10, R1-R10, and S1-S10.
3. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions P11-P20, Q11-Q20, R11-R20, and S11-S20.
4. R3 = 1.5 kΩ ±1%, 1/4 W.

* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV and XVI.

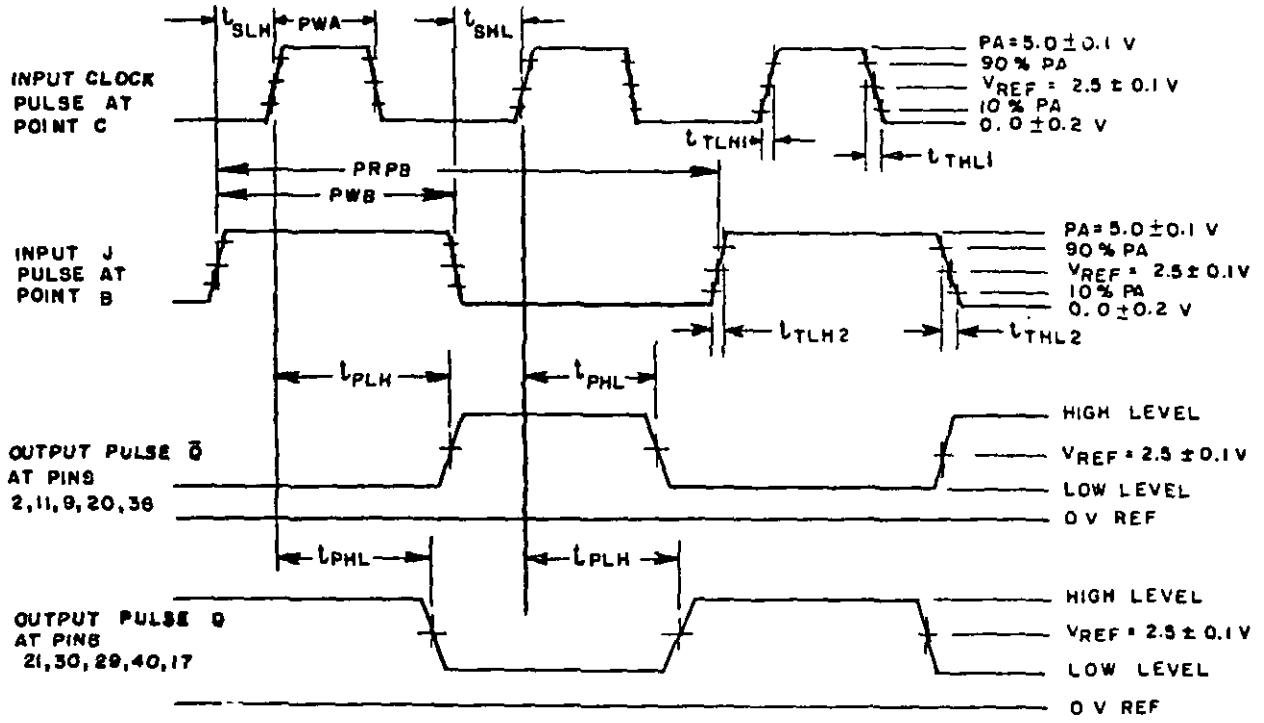
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NOTES:
1. Input pulse parameters for item J01-2:
   Put item J01-2 in the double pulse mode.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs.
2. Input pulse parameters for item J01-1:
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.
   PWB = 1.0 ±0.1 µs.
   PRPB = 4.0 ±0.4 µs.
3. $t_{HOLD} = 0.+2$ ns, -0 ns.
   $t_{DELAY} = 1.0 \pm 0.1$ µs.
4. Verify that all outputs have a high level $\geq$ 4.6 V and a low level $\leq$ 0.4 V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XV.

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NOTES:
1. Input pulse parameters for item J01-2:
   Put item J01-2 in the double pulse mode.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs.
2. Input pulse parameters for item J01-1:
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
3. $t_{SLH} = t_{SHL}$ = 200 ns ±5%.
4. Verify that all outputs have a high level ≥ 4.6 V and a low level ≤ 0.4 V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 16. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T1	1Q Propagation delay time, C_L = 50 pF (set to output), V_{DD} = 5 V	t_{PLH}	31,13,8,26, 19,37,35,14	3	24	B	21	35	350	ns	17,18
U1	$1\overline{Q}$ "	t_{PHL}	31,13,8,26, 19,37,35,14	3	24	"	2	"	"	"	"
T2	2Q "	t_{PLH}	3,24,8,26, 19,37,35,14	31	13	"	30	"	"	"	"
U2	$2\overline{Q}$ "	t_{PHL}	3,24,8,26, 19,37,35,14	31	13	"	11	"	"	"	"
T3	3Q "	t_{PLH}	3,24,31,13, 19,37,35,14	8	26	"	29	"	"	"	"
U3	$3\overline{Q}$ "	t_{PHL}	3,24,31,13, 19,37,35,14	8	26	"	9	"	"	"	"
T4	4Q "	t_{PLH}	3,24,31,13, 8,26,35,14	19	37	"	40	"	"	"	"
U4	$4\overline{Q}$ "	t_{PHL}	3,24,31,13, 8,26,35,14	19	37	"	20	"	"	"	"
T5	5Q "	t_{PLH}	3,24,31,13, 8,26,19,37	35	14	"	17	"	"	"	"
U5	$5\overline{Q}$ "	t_{PHL}	3,24,31,13, 8,26,19,37	35	14	"	36	"	"	"	"
U6	1Q Propagation delay time, C_L = 50 pF (reset to output), V_{DD} = 5 V	t_{PHL}	31,13,8,26, 19,37,35,14	3	24	C	21	45	450	"	"
T6	$1\overline{Q}$ "	t_{PLH}	31,13,8,26, 19,37,35,14	3	24	"	2	"	"	"	"
U7	2Q "	t_{PHL}	3,24,8,26, 19,37,35,14	31	13	"	30	"	"	"	"
T7	$2\overline{Q}$ "	t_{PLH}	3,24,8,26, 19,37,35,14	31	13	"	11	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

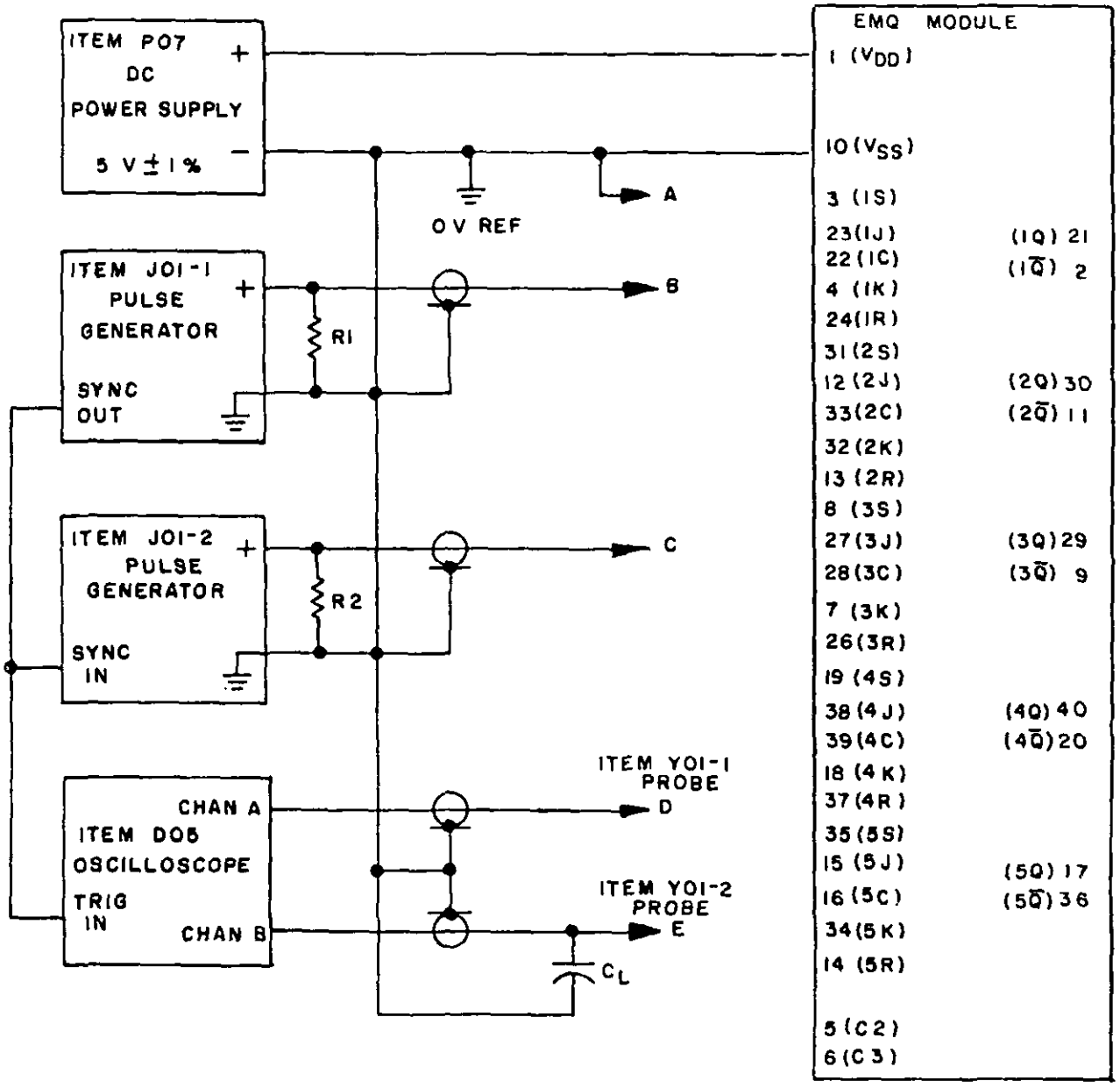
Test condition	Parameter			Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
U8	3Q	Propagation delay time, C_L = 50 pF (reset to output), V_{DD} = 5 V	t_{PHL}	3,24,31,13, 19,37,35,14	8	26	C	29	45	450	ns	17,18
T8	$3\overline{Q}$	"	t_{PLH}	3,24,31,13, 19,37,35,14	8	26	"	9	"	"	"	"
U9	4Q	"	t_{PHL}	3,24,31,13, 8,26,35,14	19	37	"	40	"	"	"	"
T9	$4\overline{Q}$	"	t_{PLH}	3,24,31,13, 8,26,35,14	19	37	"	20	"	"	"	"
U10	5Q	"	t_{PHL}	3,24,31,13, 8,26,19,37	35	14	"	17	"	"	"	"
T10	$5\overline{Q}$	"	t_{PLH}	3,24,31,13, 8,26,19,37	35	14	"	36	"	"	"	"
T11	1Q	Propagation delay time, C_L = 150 pF (set to output), V_{DD} = 5 V	t_{PLH}	31,13,8,26, 19,37,35,14	3	24	B	21	55	550	"	"
U11	$1\overline{Q}$	"	t_{PHL}	31,13,8,26, 19,37,35,14	3	24	"	2	"	"	"	"
T12	2Q	"	t_{PLH}	3,24,8,26, 19,37,35,14	31	13	"	30	"	"	"	"
U12	$2\overline{Q}$	"	t_{PHL}	3,24,8,26, 19,37,35,14	31	13	"	11	"	"	"	"
T13	3Q	"	t_{PLH}	3,24,31,13, 19,37,35,14	8	26	"	29	"	"	"	"
U13	$3\overline{Q}$	"	t_{PHL}	3,24,31,13, 19,37,35,14	8	26	"	9	"	"	"	"
T14	4Q	"	t_{PLH}	3,24,31,13, 8,26,35,14	19	37	"	40	"	"	"	"
U14	$4\overline{Q}$	"	t_{PHL}	3,24,31,13, 8,26,35,14	19	37	"	20	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T15	5Q Propagation delay time, C_L = 150 pF (set to output), V_{DD} = 5 V	t_{PLH}	3,24,31,13, 8,26,19,37	35	14	B	17	55	550	ns	17,18
U15	$5\overline{Q}$ Propagation delay time, C_L = 150 pF (set to output), V_{DD} = 5 V	t_{PHL}	3,24,31,13, 8,26,19,37	35	14	B	36	55	550	"	"
U16	1Q Propagation delay time, C_L = 150 pF (reset to output), V_{DD} = 5 V	t_{PHL}	31,13,8,26, 19,37,35,14	3	24	C	21	65	650	"	"
T16	$1\overline{Q}$ "	t_{PLH}	31,13,8,26, 19,37,35,14	3	24	"	2	"	"	"	"
U17	2Q "	t_{PHL}	3,24,8,26, 19,37,35,14	31	13	"	30	"	"	"	"
T17	$2\overline{Q}$ "	t_{PLH}	3,24,8,26, 19,37,35,14	31	13	"	11	"	"	"	"
U18	3Q "	t_{PHL}	3,24,31,13, 19,37,35,14	8	26	"	29	"	"	"	"
T18	$3\overline{Q}$ "	t_{PLH}	3,24,31,13, 19,37,35,14	8	26	"	9	"	"	"	"
U19	4Q "	t_{PHL}	3,24,31,13, 8,26,35,14	19	37	"	40	"	"	"	"
T19	$4\overline{Q}$ "	t_{PLH}	3,24,31,13, 8,26,35,14	19	37	"	20	"	"	"	"
U20	5Q "	t_{PHL}	3,24,31,13, 8,26,19,37	35	14	"	17	"	"	"	"
T20	$5\overline{Q}$ "	t_{PLH}	3,24,31,13, 8,26,19,37	35	14	"	36	"	"	"	"

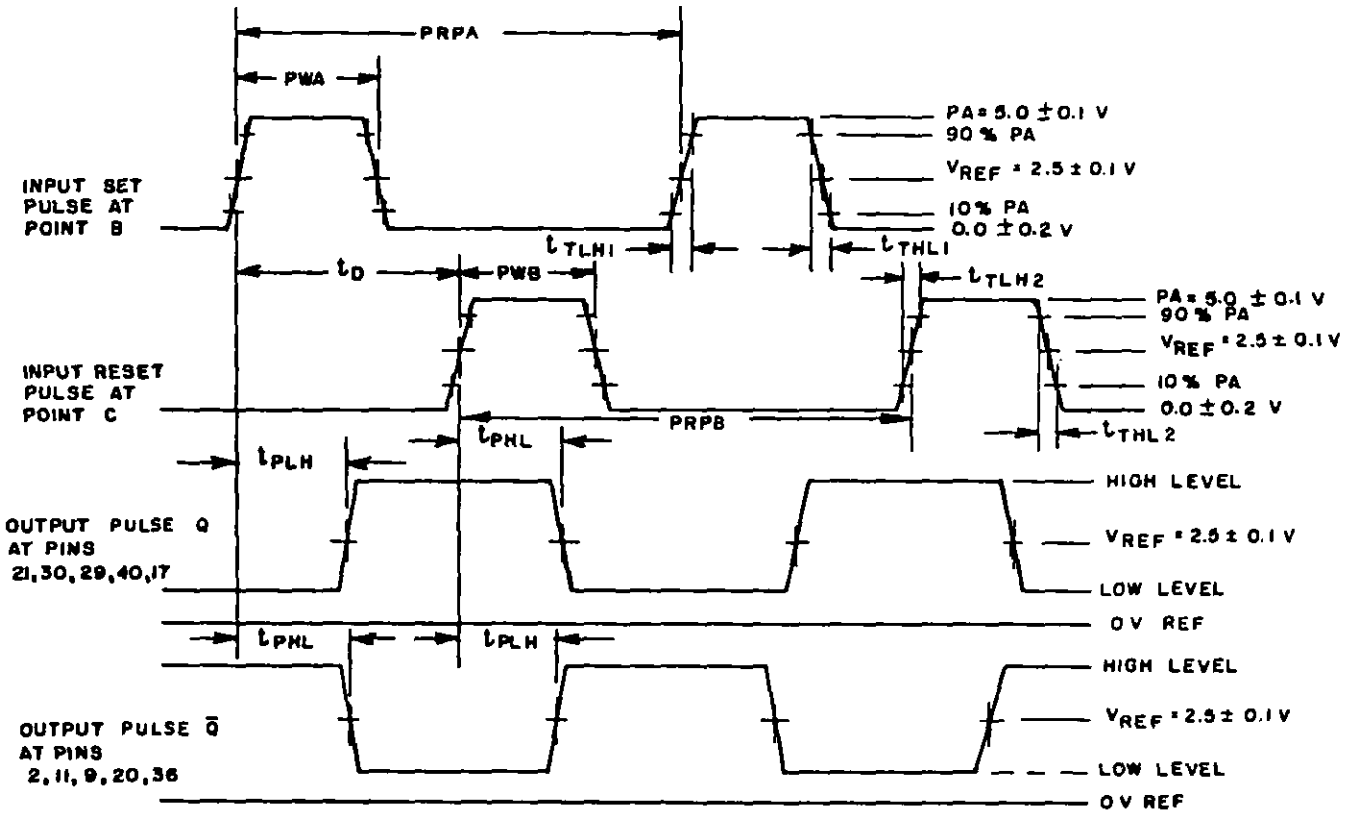
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions T1-T10 and U1-U10.
3. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions T11-T20 and U11-U20.

* FIGURE 17. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVII.

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NOTES:
1. Input pulse parameters for item J01-1:
   PRPA = 4.0 ±0.4 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   PWA = 250 ±20 ns.
2. Input pulse parameters for item J01-2:
   PRPB = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
   PWB = 250 ±20 ns.
3. t_D = 2.0 ±0.2 µs.
4. Verify that all outputs have a high level ≥ 4.6 V and a low level ≤ 0.4 V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 18. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVII.

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TABLE XVIII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter			Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1Q	Transition time, C_L = 50 pF, V_{DD} = 10 V	t_{TLH}	12,33,27,28,38, 39,15,16	23	22	21	18	180	ns	19,20
W1	$1\overline{Q}$	"	t_{THL}	12,33,27,28,38, 39,15,16	23	22	2	10	100	"	"
V2	2Q	"	t_{TLH}	23,22,27,28,38, 39,15,16	12	33	30	18	180	"	"
W2	$2\overline{Q}$	"	t_{THL}	23,22,27,28,38, 39,15,16	12	33	11	10	100	"	"
V3	3Q	"	t_{TLH}	23,22,12,33,38, 39,15,16	27	28	29	18	180	"	"
W3	$3\overline{Q}$	"	t_{THL}	23,22,12,33,38, 39,15,16	27	28	9	10	100	"	"
V4	4Q	"	t_{TLH}	23,22,12,33,27, 28,15,16	38	39	40	18	180	"	"
W4	$4\overline{Q}$	"	t_{THL}	23,22,12,33,27, 28,15,16	38	39	20	10	100	"	"
V5	5Q	"	t_{TLH}	23,22,12,33,27, 28,38,39	15	16	17	18	180	"	"
W5	$5\overline{Q}$	"	t_{THL}	23,22,12,33,27, 28,38,39	15	16	36	10	100	"	"
W6	1Q	"	t_{THL}	12,33,27,28,38, 39,15,16	23	22	21	10	100	"	"
V6	$1\overline{Q}$	"	t_{TLH}	12,33,27,28,38, 39,15,16	23	22	2	18	180	"	"
W7	2Q	"	t_{THL}	23,22,27,28,38, 39,15,16	12	33	30	10	100	"	"
V7	$2\overline{Q}$	"	t_{TLH}	23,22,27,28,38, 39,15,16	12	33	11	18	180	"	"

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TABLE XVIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter			Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
W8	3Q	Transition time, C_L = 50 pF, V_{DD} = 10 V	t_{THL}	23,22,12,33,38, 39,15,16	27	28	29	10	100	ns	19,20
V8	$3\overline{Q}$	"	t_{TLH}	23,22,12,33,38, 39,15,16	27	28	9	18	180	"	"
W9	4Q	"	t_{THL}	23,22,12,33,27, 28,15,16	38	39	40	10	100	"	"
V9	$4\overline{Q}$	"	t_{TLH}	23,22,12,33,27, 28,15,16	38	39	20	18	180	"	"
W10	5Q	"	t_{THL}	23,22,12,33,27, 28,38,39	15	16	17	10	100	"	"
V10	$5\overline{Q}$	"	t_{TLH}	23,22,12,33,27, 28,38,39	15	16	36	18	180	"	"

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TABLE XIX. *Propagation delay time (t_{PLH} and t_{PHL}) test.*

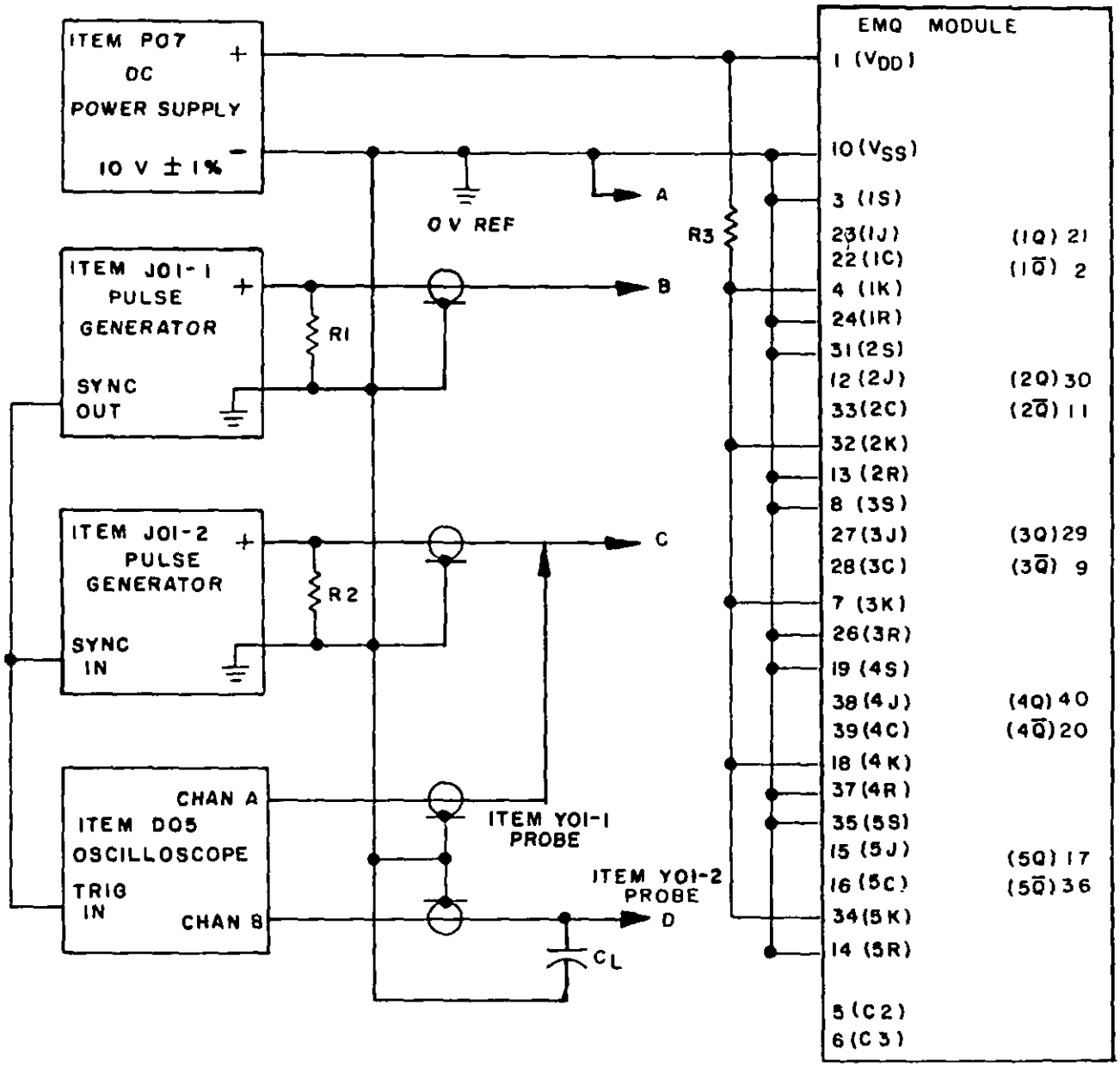
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	1Q Propagation delay time, C_L = 50 pF, (clock to output), V_{DD} = 10 V	t_{PLH}	12,33,27,28,38, 39,15,16	23	22	21	15	150	ns	19,21
Y1	$1\overline{Q}$ "	t_{PHL}	12,33,27,28,38, 39,15,16	23	22	2	"	"	"	"
X2	2Q "	t_{PLH}	23,22,27,28,38, 39,15,16	12	33	30	"	"	"	"
Y2	$2\overline{Q}$ "	t_{PHL}	23,22,27,28,38, 39,15,16	12	33	11	"	"	"	"
X3	3Q "	t_{PLH}	23,22,12,33,38, 39,15,16	27	28	29	"	"	"	"
Y3	$3\overline{Q}$ "	t_{PHL}	23,22,12,33,38, 39,15,16	27	28	9	"	"	"	"
X4	4Q "	t_{PLH}	23,22,12,33,27, 28,15,16	38	39	40	"	"	"	"
Y4	$4\overline{Q}$ "	t_{PHL}	23,22,12,33,27, 28,15,16	38	39	20	"	"	"	"
X5	5Q "	t_{PLH}	23,22,12,33,27, 28,38,39	15	16	17	"	"	"	"
Y5	$5\overline{Q}$ "	t_{PHL}	23,22,12,33,27, 28,38,39	15	16	36	"	"	"	"
Y6	1Q "	t_{PHL}	12,33,27,28,38, 39,15,16	23	22	21	"	"	"	"
X6	$1\overline{Q}$ "	t_{PLH}	12,33,27,28,38, 39,15,16	23	22	2	"	"	"	"
Y7	2Q "	t_{PHL}	23,22,27,28,38, 39,15,16	12	33	30	"	"	"	"
X7	$2\overline{Q}$ "	t_{PLH}	23,22,27,28,38, 39,15,16	12	33	11	"	"	"	"

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TABLE XIX. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Y8	3Q Propagation delay time, C_L = 50 pF, (clock to output), V_{DD} = 10 V	t_{PHL}	23,22,12,33,38, 39,15,16	27	28	29	15	150	ns	19,21
X8	$3\overline{Q}$	t_{PLH}	23,22,12,33,38, 39,15,16	27	28	9	"	"	"	"
Y9	4Q	t_{PHL}	23,22,12,33,27, 28,15,16	38	39	40	"	"	"	"
X9	$4\overline{Q}$	t_{PLH}	23,22,12,33,27, 28,15,16	38	39	20	"	"	"	"
Y10	5Q	t_{PHL}	23,22,12,33,27, 28,38,39	15	16	17	"	"	"	"
X10	$5\overline{Q}$	t_{PLH}	23,22,12,33,27, 28,38,39	15	16	36	"	"	"	"

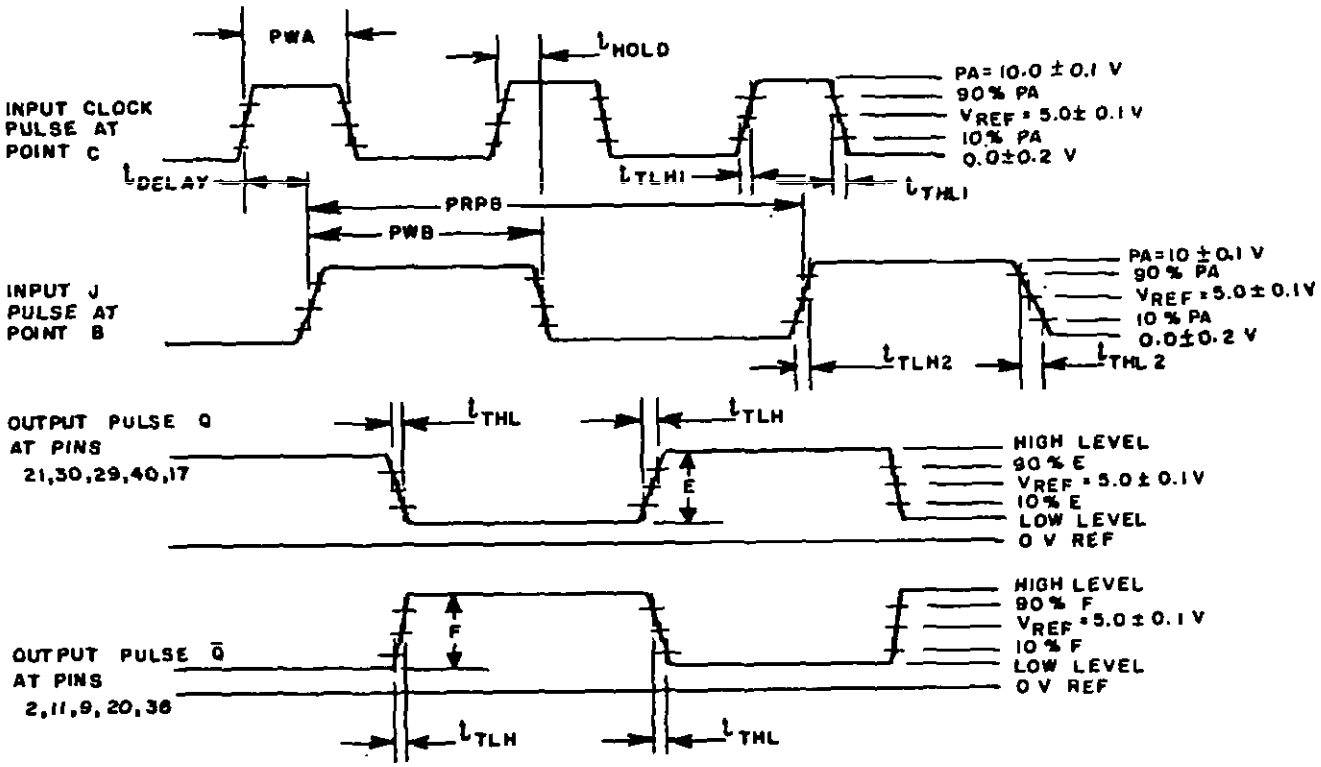
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:13Z
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 2 W.
2. R3 = 1.5 kΩ ±1%, 1/4 W.
3. C_L = 50 ±5 pF (including probe and parasitic capacitance).

* FIGURE 19. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVIII and XIX.

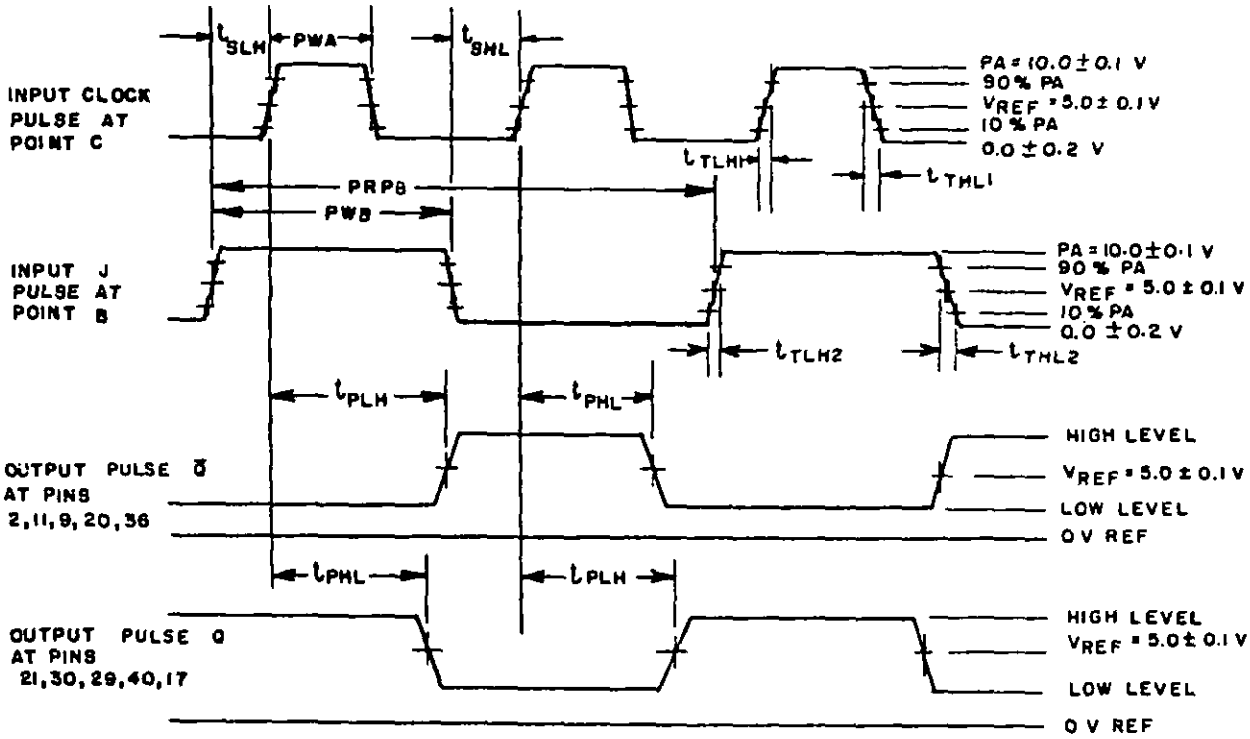
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NOTES:
1. Input pulse parameters for item J01-2:
   Put item J01-2 in the double pulse mode.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs.
2. Input pulse parameters for item J01-1:
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
3. $t_{HOLD} = 0$ +2 ns, -0 ns.
   $t_{DELAY} = 1.0 \pm 0.1$ µs.
4. Verify that all outputs have a high level $\geq$ 9.5 V and a low level $\leq$ 0.5 V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 20. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XVIII.

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NOTES:
1. Input pulse parameters for item J01-2:
   Put item J01-2 in the double pulse mode.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs.
2. Input pulse parameters for item J01-1:
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
3. $t_{SLH} = t_{SHL} = 200$ ns ±5%.
4. Verify that all outputs have a high level ≥ 9.5 V and a low level ≤ 0.5 V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIX.

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TABLE XX. Propagation delay time (t_{PLH} and t_{PHL}) test.

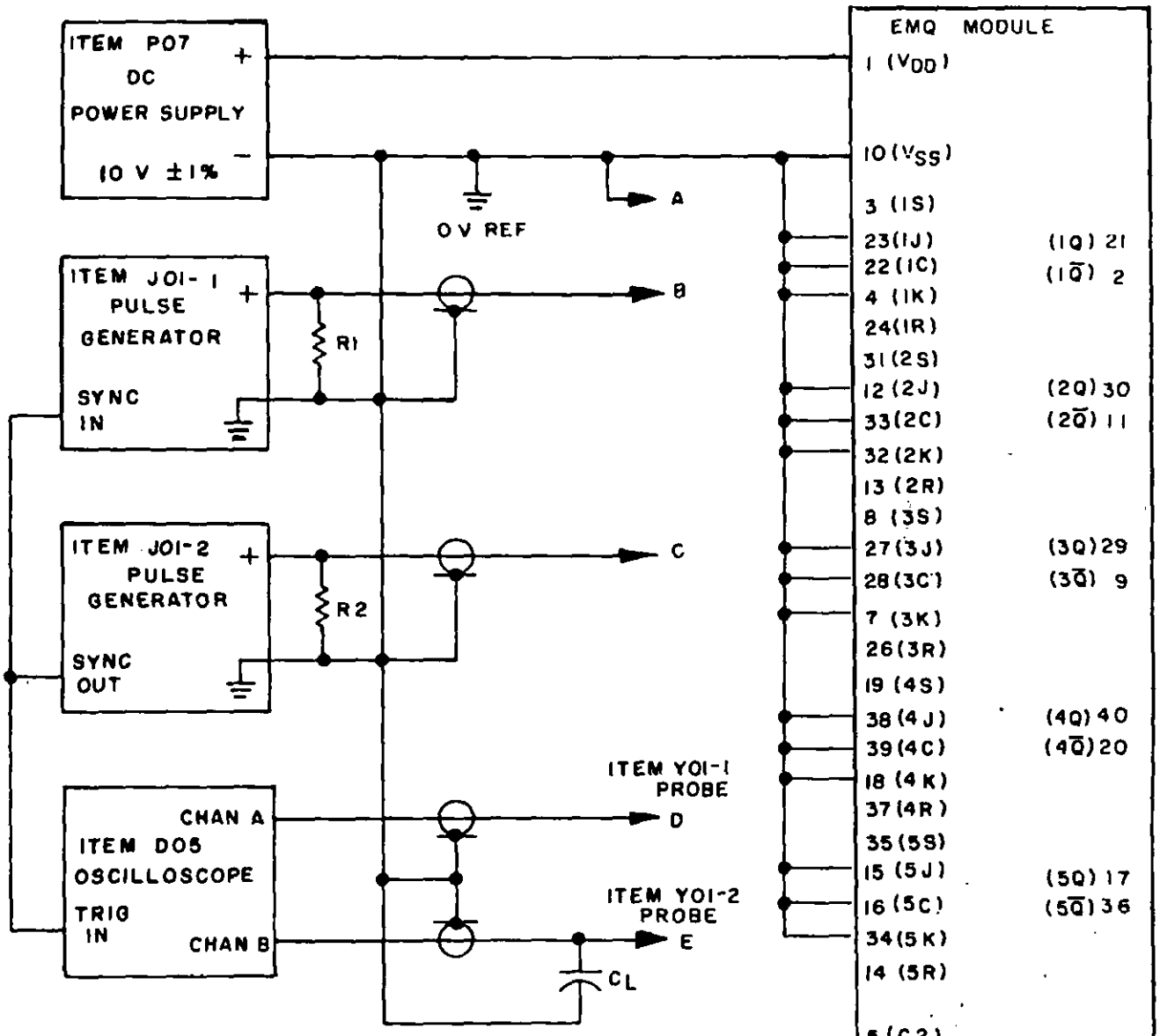
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z1	1Q Propagation delay time, C_L = 50 pF (set to output), V_{DD} = 10 V	t_{PLH}	31,13,8,26,19,37, 35,14	3	24	8	21	15	150	ns	22,23
AA1	$1\overline{Q}$ "	t_{PHL}	31,13,8,26,19,37, 35,14	3	24	"	2	"	"	"	"
Z2	2Q "	t_{PLH}	3,24,8,26,19,37, 35,14	31	13	"	30	"	"	"	"
AA2	$2\overline{Q}$ "	t_{PHL}	3,24,8,26,19,37, 35,14	31	13	"	11	"	"	"	"
Z3	3Q "	t_{PLH}	3,24,31,13,19,37, 35,14	8	26	"	29	"	"	"	"
AA3	$3\overline{Q}$ "	t_{PHL}	3,24,31,13,19,37, 35,14	8	26	"	9	"	"	"	"
Z4	4Q "	t_{PLH}	3,24,31,13,8,26, 35,14	19	37	"	40	"	"	"	"
AA4	$4\overline{Q}$ "	t_{PHL}	3,24,31,13,8,26, 35,14	19	37	"	20	"	"	"	"
Z5	5Q "	t_{PLH}	3,24,31,13,8,26, 19,37	35	14	"	17	"	"	"	"
AA5	$5\overline{Q}$ "	t_{PHL}	3,24,31,13,8,26, 19,37	35	14	"	36	"	"	"	"
AA6	1Q Propagation delay time, C_L = 50 pF (reset to output), V_{DD} = 10 V	t_{PHL}	31,13,8,26,19,37, 35,14	3	24	C	21	20	200	"	"
Z6	$1\overline{Q}$ "	t_{PLH}	31,13,8,26,19,37, 35,14	3	24	"	2	"	"	"	"
AA7	2Q "	t_{PHL}	3,24,8,26,19,37, 35,14	31	13	"	30	"	"	"	"
Z7	$2\overline{Q}$ "	t_{PLH}	3,24,8,26,19,37, 35,14	31	13	"	11	"	"	"	"

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TABLE XX. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA8	3Q Propagation delay time, C_L = 50 pF (reset to output), V_{DD} = 10 V	t_{PHL}	3,24,31,13,19,37, 35,14	8	26	C	29	20	200	ns	22,23
Z8	$3\overline{Q}$ "	t_{PLH}	3,24,31,13,19,37, 35,14	8	26	"	9	"	"	"	"
AA9	4Q "	t_{PHL}	3,24,31,13,8,26, 35,14	19	37	"	40	"	"	"	"
Z9	$4\overline{Q}$ "	t_{PLH}	3,24,31,13,8,26, 35,14	19	37	"	20	"	"	"	"
AA10	5Q "	t_{PHL}	3,24,31,13,8,26, 19,37	35	14	"	17	"	"	"	"
Z10	$5\overline{Q}$ "	t_{PLH}	3,24,31,13,8,26, 19,37	35	14	"	36	"	"	"	"

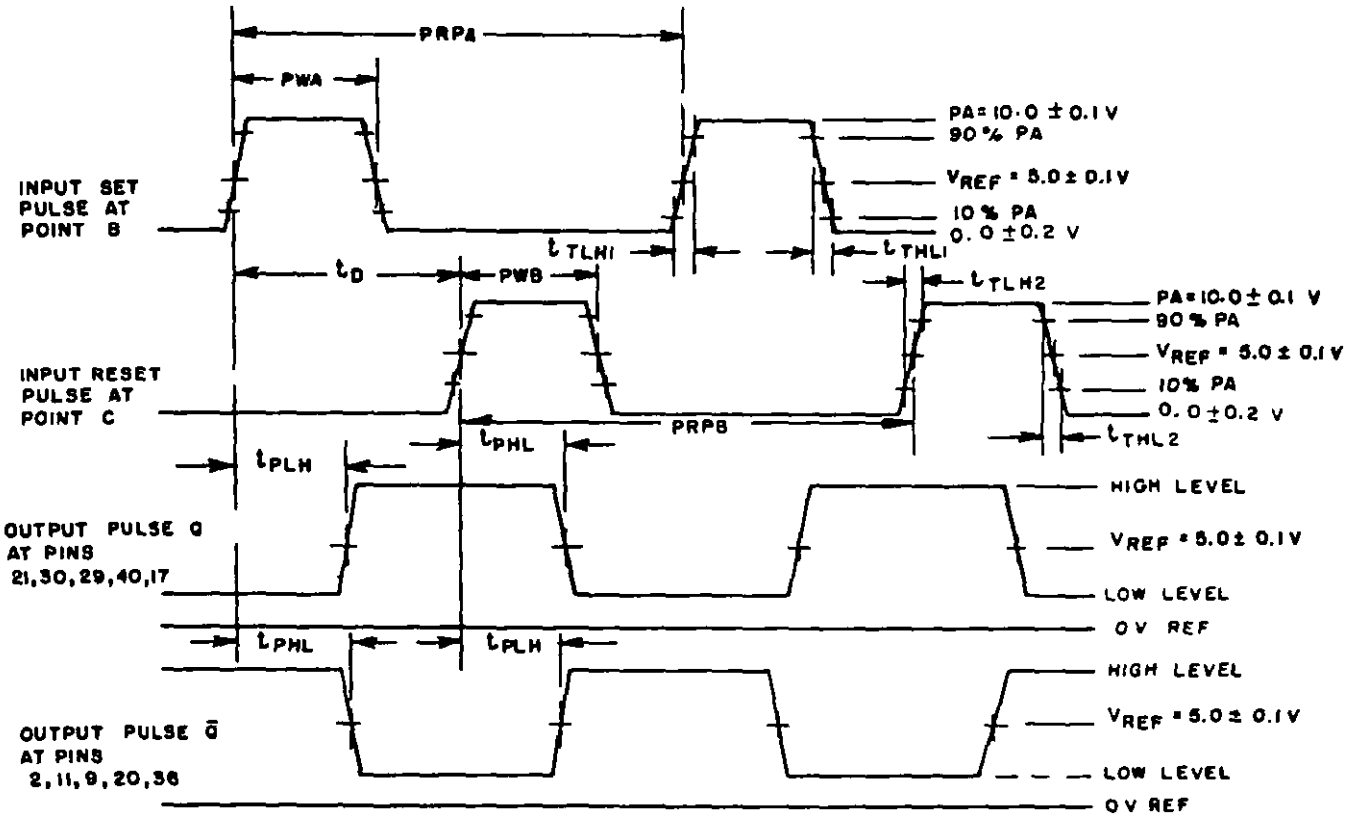
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance).

* FIGURE 22. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XX.

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NOTES:
1. Input pulse parameters for item J01-1:
   PRPA = 4.0 ±0.4 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   PWA = 250 ±20 ns.
2. Input pulse parameters for item J01-2:
   PRPB = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
   PWB = 250 ±20 ns.
3. t_D = 2.0 ±0.2 µs.
4. Verify that all outputs have a high level ≥ 9.6 V and a low level ≤ 0.5 V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 23. Propagation delay time (t_{TLH} and t_{THL}) test circuit waveforms for table XX.

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TABLE XXI. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
CB1	1Q Transition time, C_L = 50 pF, V_{DD} = 15 V	t_{TLH}	12,33,27,28,38, 39,15,16	23	22	21	13	130	ns	24,25
CC1	$1\overline{Q}$ "	t_{THL}	12,33,27,28,38, 39,15,16	23	22	2	8	80	"	"
BB2	2Q "	t_{TLH}	23,22,27,28,38, 39,15,16	12	33	30	13	130	"	"
CC2	$2\overline{Q}$ "	t_{THL}	23,22,27,28,38, 39,15,16	12	33	11	8	80	"	"
BB3	3Q "	t_{TLH}	23,22,12,33,38, 39,15,16	27	28	29	13	130	"	"
CC3	$3\overline{Q}$ "	t_{THL}	23,22,12,33,38, 39,15,16	27	28	9	8	80	"	"
BB4	4Q "	t_{TLH}	23,22,12,33,27, 28,15,16	38	39	40	13	130	"	"
CC4	$4\overline{Q}$ "	t_{THL}	23,22,12,33,27, 28,15,16	38	39	20	8	80	"	"
BB5	5Q "	t_{TLH}	23,22,12,33,27, 28,38,39	15	16	17	13	130	"	"
CC5	$5\overline{Q}$ "	t_{THL}	23,22,12,33,27, 28,38,39	15	16	36	8	80	"	"
CC6	1Q "	t_{THL}	12,33,27,28,38, 39,15,16	23	22	21	8	80	"	"
BB6	$1\overline{Q}$ "	t_{TLH}	12,33,27,28,38, 39,15,16	23	22	2	13	130	"	"
CC7	2Q "	t_{THL}	23,22,27,28,38, 39,15,16	12	33	30	8	80	"	"
BB7	$2\overline{Q}$ "	t_{TLH}	23,22,27,28,38, 39,15,16	12	33	11	13	130	"	"

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TABLE XXI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
CC8	3Q Transition time, C_L = 50 pF, V_{DD} = 15 V	t_{THL}	23,22,12,33,38, 39,15,16	27	28	29	8	80	ns	24,25
BB8	$3\overline{Q}$ "	t_{TLH}	23,22,12,33,38, 39,15,16	27	28	9	13	130	"	"
CC9	4Q "	t_{THL}	23,22,12,33,27, 28,15,16	38	39	40	8	80	"	"
BB9	$4\overline{Q}$ "	t_{TLH}	23,22,12,33,27, 28,15,16	38	39	20	13	130	"	"
CC10	5Q "	t_{THL}	23,22,12,33,27, 28,38,39	15	16	17	8	80	"	"
BB10	$5\overline{Q}$ "	t_{TLH}	23,22,12,33,27, 28,38,39	15	16	36	13	130	"	"

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TABLE XXII. Propagation delay time (t_{PLH} and t_{PHL}) test.

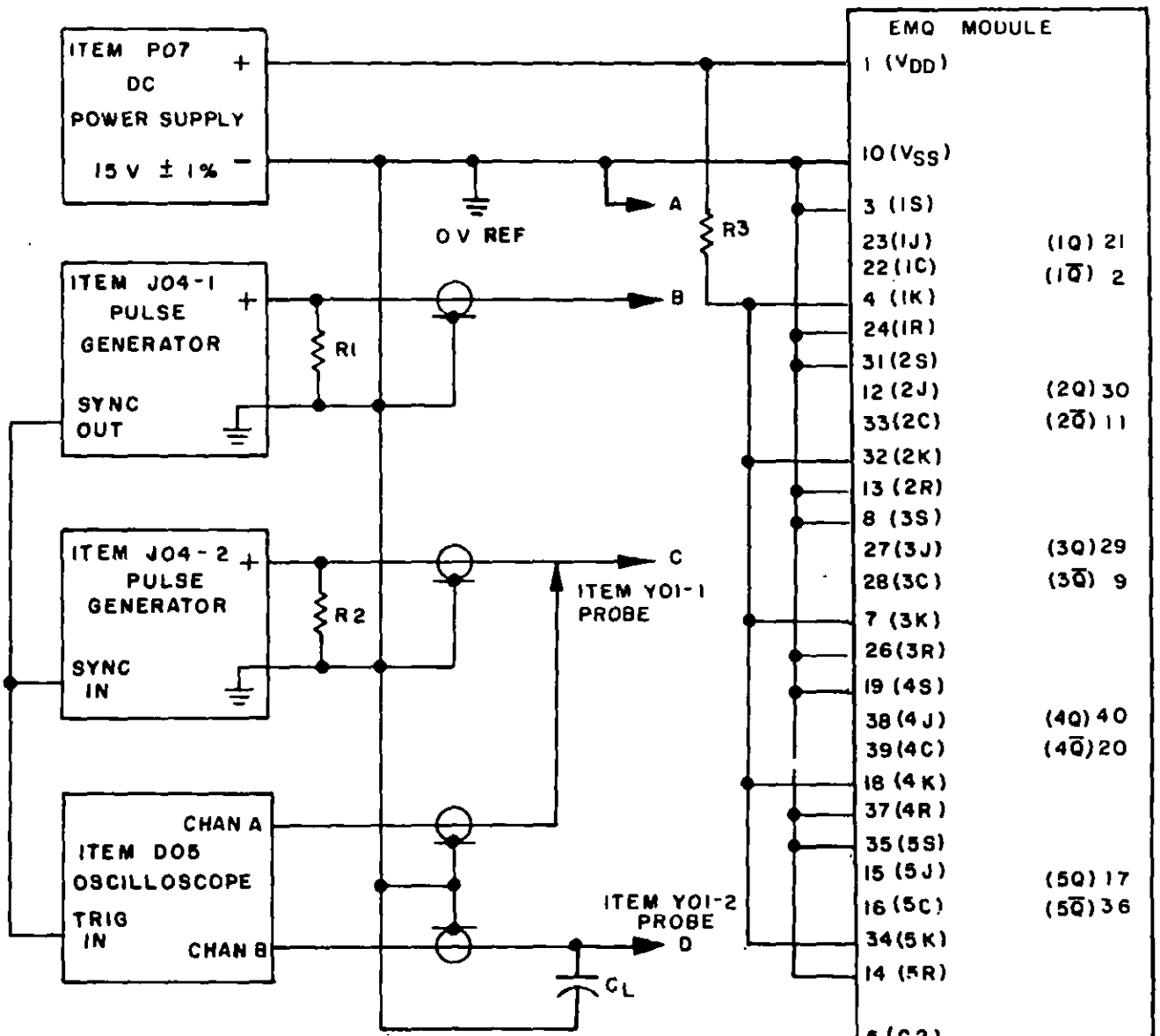
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
DD1	1Q Propagation delay time, C_L = 50 pF (clock to output), V_{DD} = 15 V	t_{PLH}	12,33,27,28,38, 39,15,16	23	22	21	10	100	ns	24,26
EE1	$1\overline{Q}$	"	t_{PHL}	12,33,27,28,38, 39,15,16	23	22	2	"	"	"	"
DD2	2Q	"	t_{PLH}	23,22,27,28,38, 39,15,16	12	33	30	"	"	"	"
EE2	$2\overline{Q}$	"	t_{PHL}	23,22,27,28,38, 39,15,16	12	33	11	"	"	"	"
DD3	3Q	"	t_{PLH}	23,22,12,33,38, 39,15,16	27	28	29	"	"	"	"
EE3	$3\overline{Q}$	"	t_{PHL}	23,22,12,33,38, 39,15,16	27	28	9	"	"	"	"
DD4	4Q	"	t_{PLH}	23,22,12,33,27, 28,15,16	38	39	40	"	"	"	"
EE4	$4\overline{Q}$	"	t_{PHL}	23,22,12,33,27, 28,15,16	38	39	20	"	"	"	"
DD5	5Q	"	t_{PLH}	23,22,12,33,27, 28,38,39	15	16	17	"	"	"	"
EE5	$5\overline{Q}$	"	t_{PHL}	23,22,12,33,27, 28,38,39	15	16	36	"	"	"	"
EE6	1Q	"	t_{PHL}	12,33,27,28,38, 39,15,16	23	22	21	"	"	"	"
DD6	$1\overline{Q}$	"	t_{PLH}	12,33,27,28,38, 39,15,16	23	22	2	"	"	"	"

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TABLE XXII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
EE7	2Q Propagation delay time, C_L = 50 pF (clock to output), V_{DD} = 15 V	t_{PHL}	23,22,27,38,28, 39,15,16	12	33	30	10	100	ns	24,26
DD7	$2\overline{Q}$ "	t_{PLH}	23,22,27,38,28, 39,15,16	12	33	11	"	"	"	"
EE8	3Q "	t_{PHL}	23,22,12,33,38, 39,15,16	27	28	29	"	"	"	"
DD8	$3\overline{Q}$ "	t_{PLH}	23,22,12,33,38, 39,15,16	27	28	9	"	"	"	"
EE9	4Q "	t_{PHL}	22,23,12,33,27, 28,15,16	38	39	40	"	"	"	"
DD9	$4\overline{Q}$ "	t_{PLH}	22,23,12,33,27, 28,15,16	38	39	20	"	"	"	"
EE10	5Q "	t_{PHL}	23,22,12,33,27, 28,38,39	15	16	17	"	"	"	"
DD10	$5\overline{Q}$ "	t_{PLH}	23,22,12,33,27, 28,38,39	15	16	36	"	"	"	"

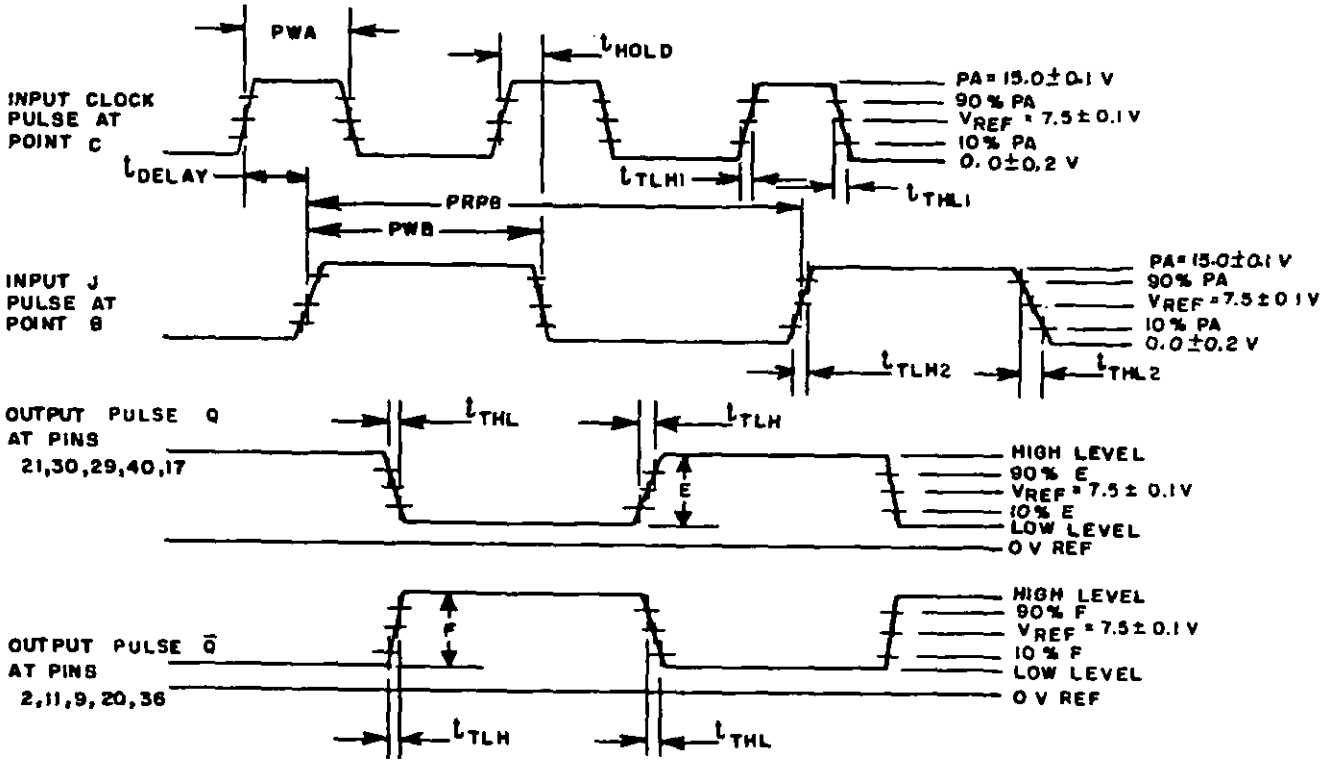
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:13Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1Ω ±5%, 2 W.
2. R3 = 1.5 kΩ ±1%, 1/4 W.
3. C_L = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 24. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXI and XXII.

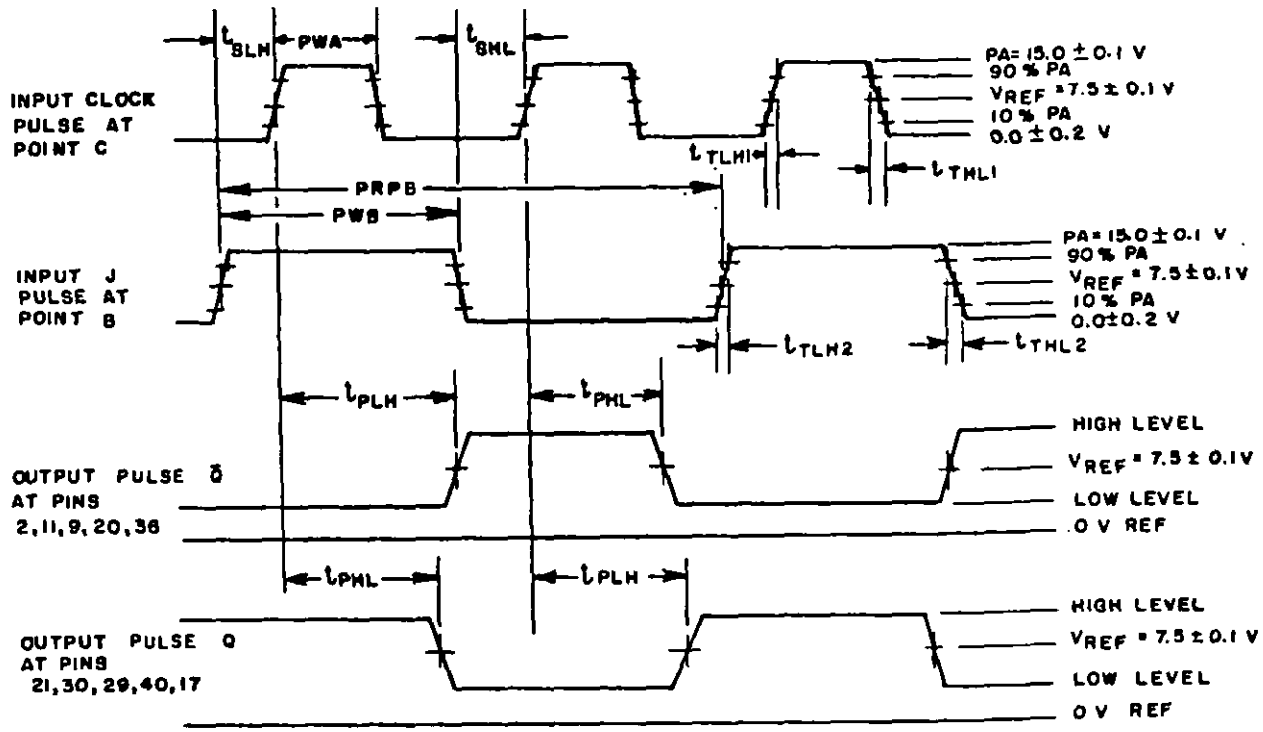
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NOTES:
1. Input pulse parameters for item J04-2:
   Put item J04-2 in the double pulse mode.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs.
2. Input pulse parameters for item J04-1:
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
3. $t_{HOLD} = 0$ +2 ns, -0 ns.
   $t_{DELAY} = 1.0 \pm 0.1$ µs.
4. Verify that all outputs have a high level ≥ 13.5 V and a low level ≤ 1.5 V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch(25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 25. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XXI.

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NOTES:
1. Input pulse parameters for item J04-2:
   Put item J04-2 in the double pulse mode.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs.
2. Input pulse parameters for item J04-1:
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
3. $t_{SLH} = t_{SHL} = 200$ ns ±5%.
4. Verify that all outputs have a high level ≥ 19.5 V and a low level ≤ 1.5 V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 26. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XXII.

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TABLE XXIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

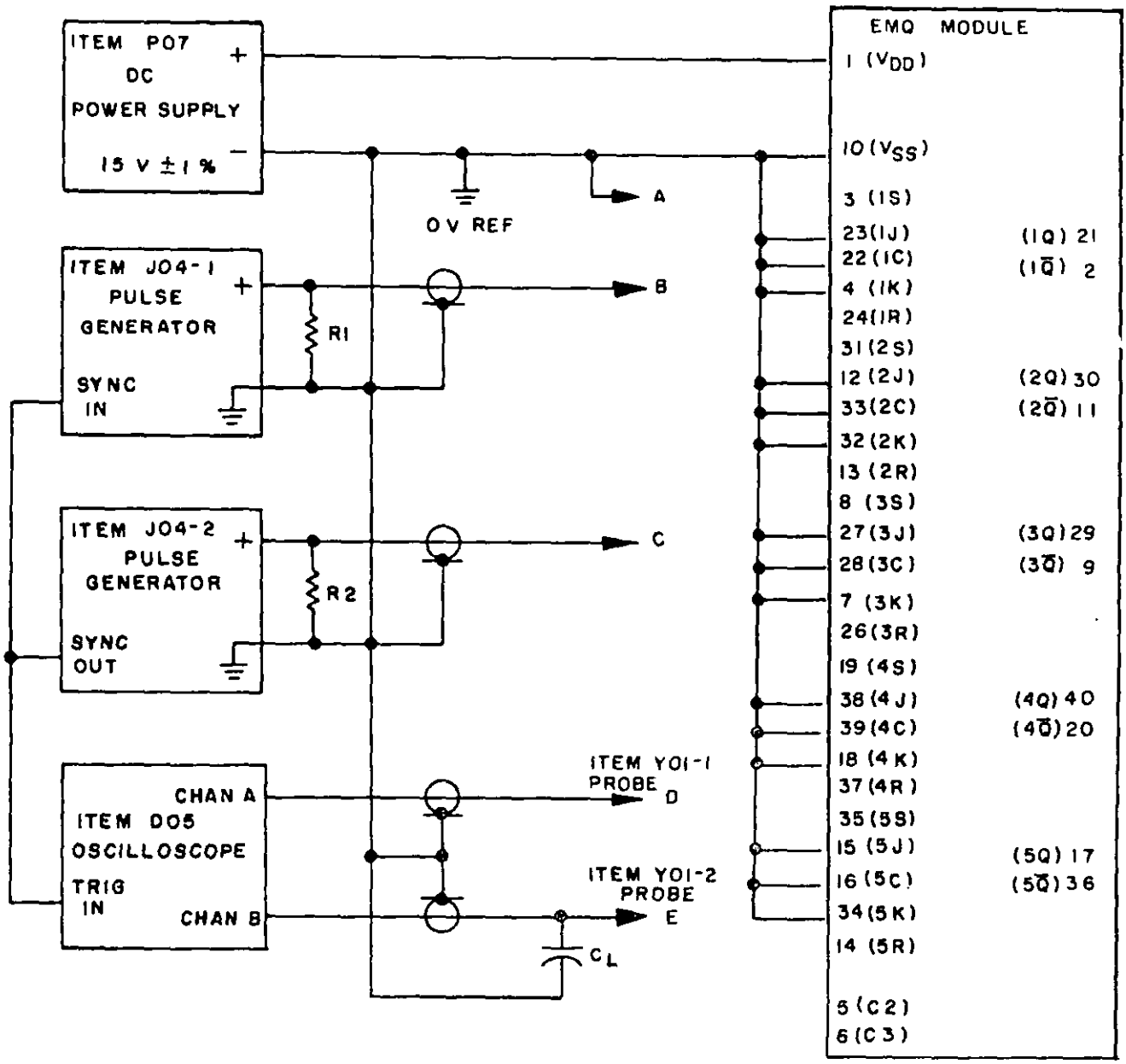
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
FF1	1Q Propagation delay time, C_L = 50 pF (set to output), V_{DD} = 15 V	t_{PLH}	31,13,8,26,19,37, 35,14	3	24	B	21	10	100	ns	27,28
GG1	$1\overline{Q}$	t_{PHL}	31,13,8,26,19,37, 35,14	3	24	"	2	"	"	"	"
FF2	2Q	t_{PLH}	3,24,8,26,19,37, 35,14	31	13	"	30	"	"	"	"
GG2	$2\overline{Q}$	t_{PHL}	3,24,8,26,19,37, 35,14	31	13	"	11	"	"	"	"
FF3	3Q	t_{PLH}	3,24,31,13,19,37, 35,14	8	26	"	29	"	"	"	"
GG3	$3\overline{Q}$	t_{PHL}	3,24,31,13,19,37, 35,14	8	26	"	9	"	"	"	"
FF4	4Q	t_{PLH}	3,24,31,13,8,26, 35,14	19	37	"	40	"	"	"	"
GG4	$4\overline{Q}$	t_{PHL}	3,24,31,13,8,26, 35,14	19	37	"	20	"	"	"	"
FF5	5Q	t_{PLH}	3,24,31,13,8,26, 19,37	35	14	"	17	"	"	"	"
GG5	$5\overline{Q}$	t_{PHL}	3,24,31,13,8,26, 19,37	35	14	"	36	"	"	"	"
GG6	1Q Propagation delay time, C_L = 50 pF (reset to output), V_{DD} = 15 V	t_{PHL}	31,13,8,26,19,37, 35,14	3	24	C	21	15	150	"	"
FF6	$1\overline{Q}$	t_{PLH}	31,13,8,26,19,37, 35,14	3	24	"	2	"	"	"	"
GG7	2Q	t_{PHL}	3,24,8,26,19,37, 35,14	31	13	"	30	"	"	"	"
FF7	$2\overline{Q}$	t_{PLH}	3,24,8,26,19,37, 35,14	31	13	"	11	"	"	"	"

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TABLE XXIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
GG8	3Q Propagation delay time, C_L = 50 pF (reset to output), V_{DD} = 15 V	t_{PHL}	3,24,31,13,19,37, 35,14	8	26	C	29	15	150	ns	27,28
FF8	$3\overline{Q}$	"	t_{PLH}	3,24,31,13,19,37, 35,14	8	26	"	9	"	"	"	"
GG9	4Q	"	t_{PHL}	3,24,31,13,8,26, 35,14	19	37	"	40	"	"	"	"
FF9	$4\overline{Q}$	"	t_{PLH}	3,24,31,13,8,26, 35,14	19	37	"	20	"	"	"	"
GG10	5Q	"	t_{PHL}	3,24,31,13,8,26, 19,37	35	14	"	17	"	"	"	"
FF10	$5\overline{Q}$	"	t_{PLH}	3,24,31,13,8,26, 19,37	35	14	"	36	"	"	"	"

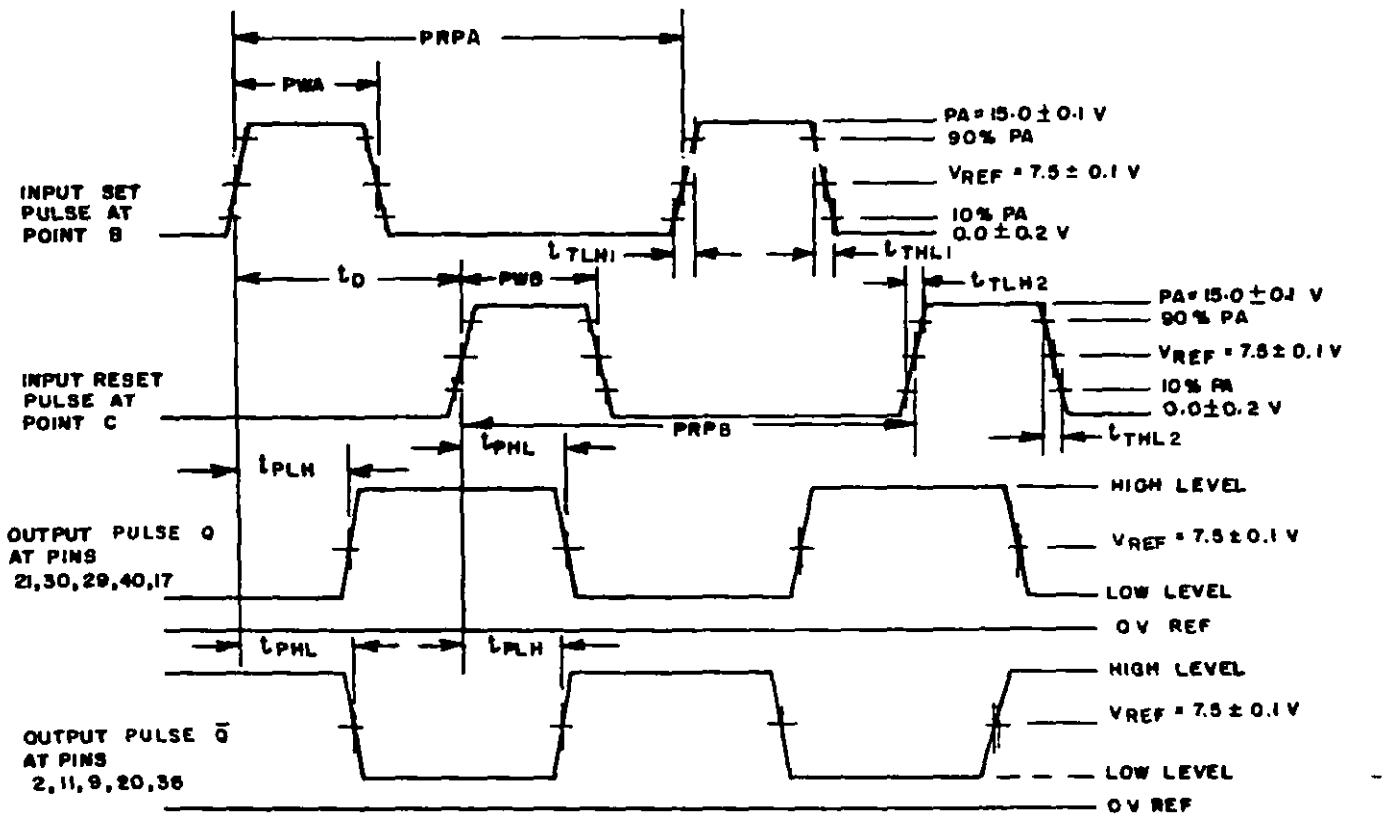
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 27. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XXIII.

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NOTES:
1. Input pulse parameters for item J04-1:
   PRPA = 4.0 ±0.4 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   PWA = 250 ±20 ns.
2. Input pulse parameters for item J04-2:
   PRPB = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
   PWB = 250 ±20 ns.
3. t_D = 2.0 ±0.2 µs.
4. Verify that all outputs have a high level ≥ 13.5 V and a low level ≤ 1.5 V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

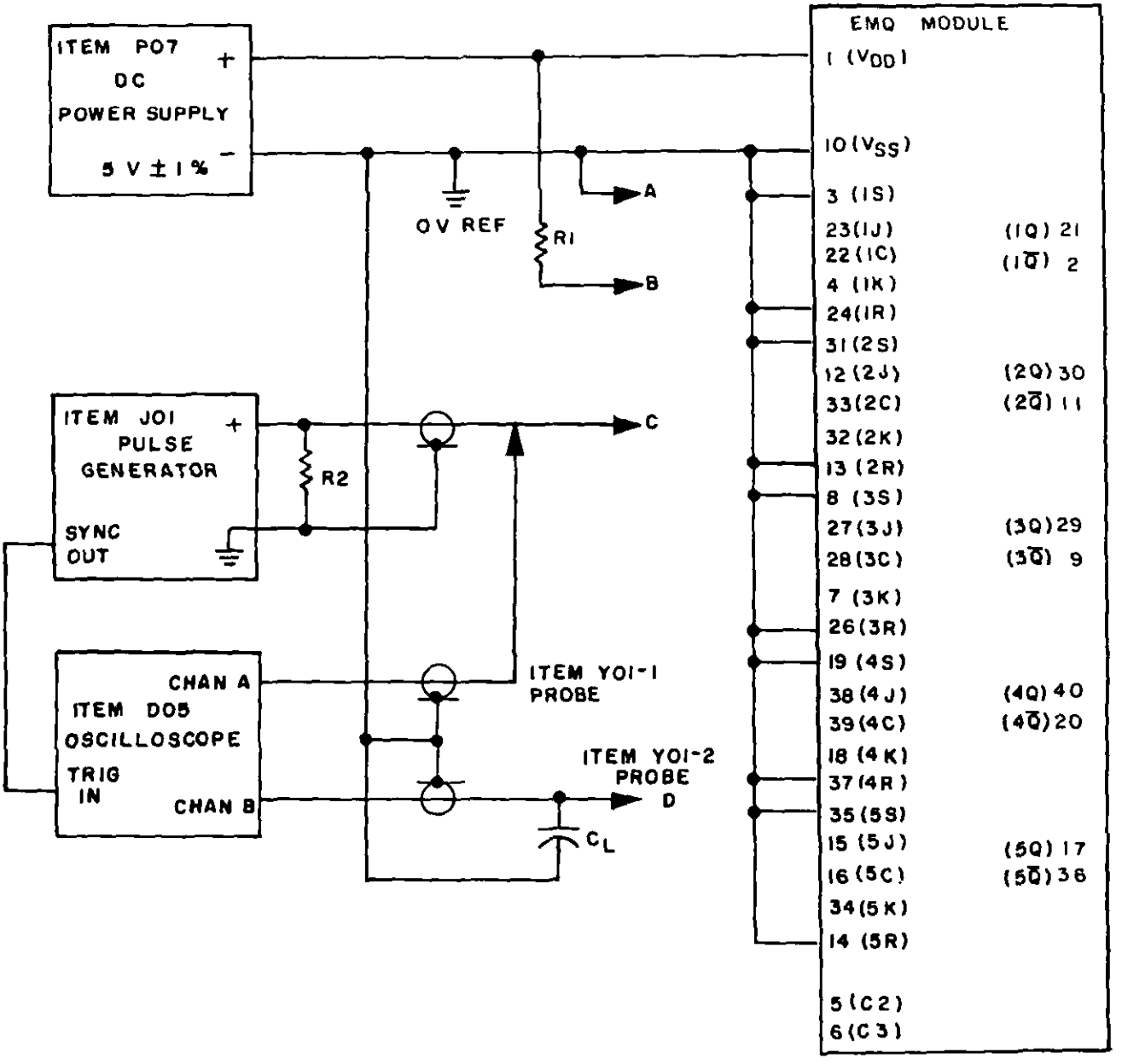
* FIGURE 28. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XXIII.

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TABLE XXIV. Maximum shift frequency (TA) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
HH1	1Q Maximum shift frequency (TA)	12,33,32,27,28,7, 38,39,18,15,16,34	23,4	22	21	1200	1465	ns	29,30
HH2	$1\overline{Q}$ "	12,33,32,27,28,7, 38,39,18,15,16,34	23,4	22	2	"	"	"	"
HH3	2Q "	23,22,4,27,28,7, 38,39,18,15,16,34	12,32	33	30	"	"	"	"
HH4	$2\overline{Q}$ "	23,22,4,27,28,7, 38,39,18,15,16,34	12,32	33	11	"	"	"	"
HH5	3Q "	23,22,4,12,33,32, 38,39,18,15,16,34	27,7	28	29	"	"	"	"
HH6	$3\overline{Q}$ "	23,22,4,12,33,32, 38,39,18,15,16,34	27,7	28	9	"	"	"	"
HH7	4Q "	23,22,4,12,33,32, 27,28,7,15,16,34	38,18	39	40	"	"	"	"
HH8	$4\overline{Q}$ "	23,22,4,12,33,32, 27,28,7,15,16,34	38,18	39	20	"	"	"	"
HH9	5Q "	23,22,4,12,33,32, 27,28,7,38,39,18	15,34	16	17	"	"	"	"
HH10	$5\overline{Q}$ "	23,22,4,12,33,32, 27,28,7,38,39,18	15,34	16	36	"	"	"	"

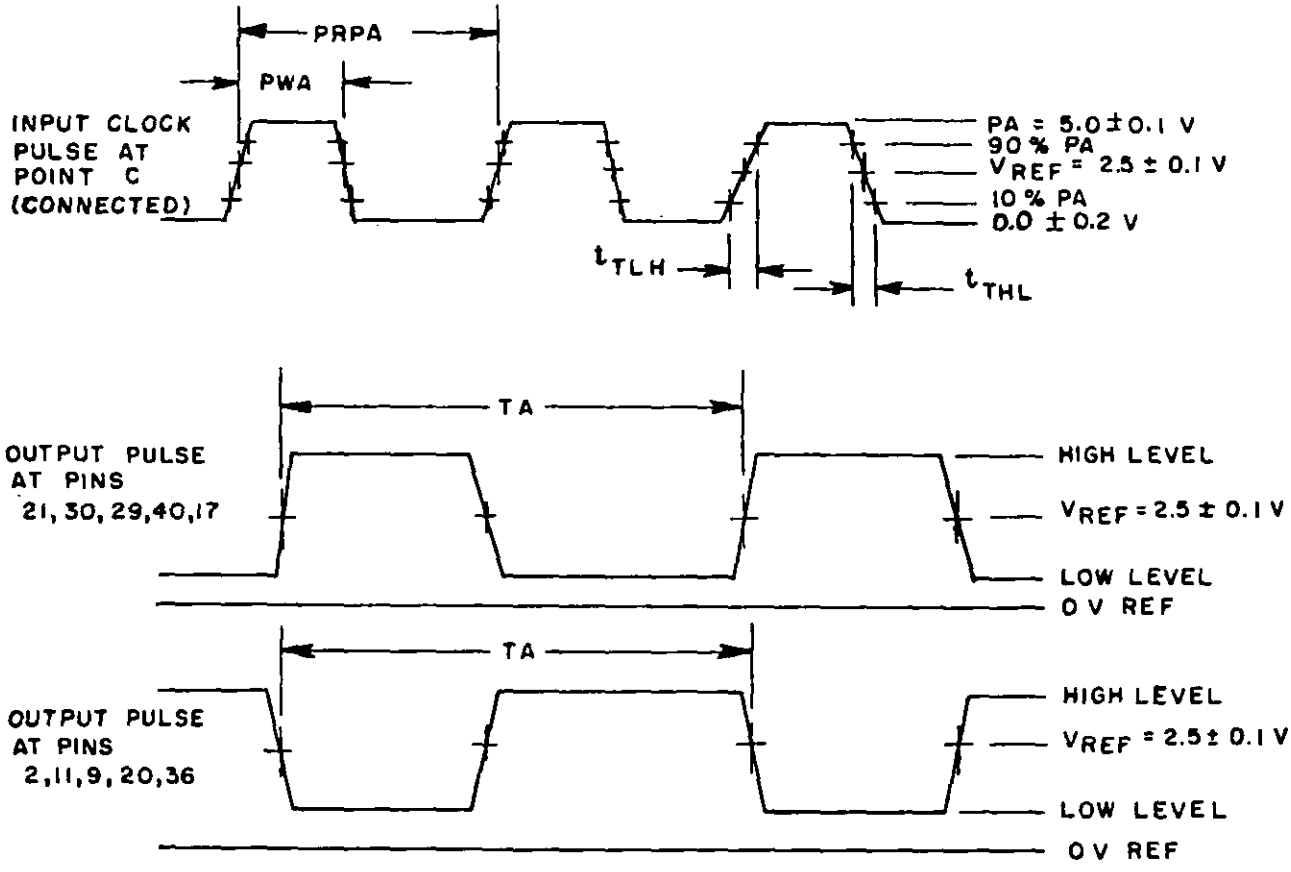
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 29. Maximum shift frequency (TA) test circuit for table XXIV.

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NOTES:
1. Input pulse parameters for item J01:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 330 +30, -0 ns.
   PRPA = 1.5 MHz ±1%.
2. Verify that the output pulse high level is $\geq$ 4.6 V and the low level is $\leq$ 0.4 V before making the measurements.
3. The lead length of electrical conductors between resistor R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between lead circuit and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 30. Maximum shift frequency test circuit waveforms for table XXIV.

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TABLE XXV. Power supply filter capacitance.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
JJ	Power supply filter capacitance	Open	1	0.007	0.013	µF	31

*TABLE XXVI. C2 and C3 capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
KK1	C2 Capacitance	5	Open	176	264	pF	31
KK2	C3 Capacitance	6	Open	176	264	pF	31

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TABLE XXVII. Input capacitance test.

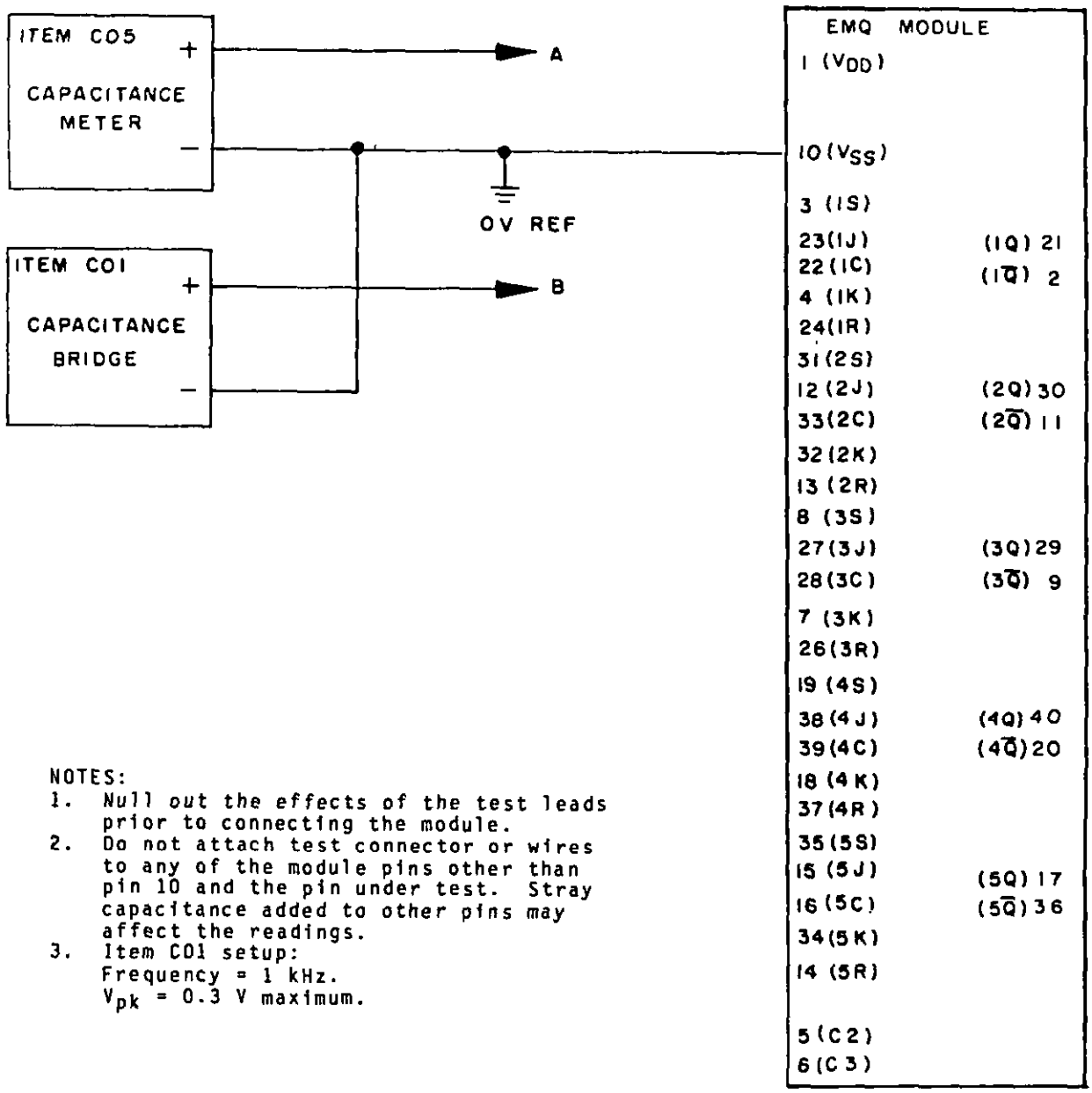
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
LL1	1S Input capacitance	3	Open	2	15	pF	31
LL2	1J "	23	"	"	"	"	"
LL3	1C "	22	"	"	"	"	"
LL4	1K "	4	"	"	"	"	"
LL5	1R "	24	"	"	"	"	"
LL6	2S "	31	"	"	"	"	"
LL7	2J "	12	"	"	"	"	"
LL8	2C "	33	"	"	"	"	"
LL9	2K "	32	"	"	"	"	"
LL10	2R "	13	"	"	"	"	"
LL11	3S "	8	"	"	"	"	"
LL12	3J "	27	"	"	"	"	"
LL13	3C "	28	"	"	"	"	"
LL14	3K "	7	"	"	"	"	"
LL15	3R "	26	"	"	"	"	"
LL16	4S "	19	"	"	"	"	"
LL17	4J "	38	"	"	"	"	"
LL18	4C "	39	"	"	"	"	"
LL19	4K "	18	"	"	"	"	"
LL20	4R "	37	"	"	"	"	"
LL21	5S "	35	"	"	"	"	"
LL22	5J "	15	"	"	"	"	"

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TABLE XXVII. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
LL23	5C Input capacitance	16	Open	2	15	pF	31
LL24	5K "	34	"	"	"	"	"
LL25	5R "	14	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item CO1 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

* FIGURE 31. Power supply filter, C2 and C3, and input capacitance test circuit for tables XXV, XXVI, and XXVII.

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